Trade and US Inequality in the Tokyo Round*

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Abstract

Against a backdrop of sharply rising inequality, the Tokyo Round of the GATT resulted in a 1.6 percentage point reduction in average US tariffs – larger than CUSFTA, NAFTA, and the liberalization accompanying the granting of PNTR to China. We construct a novel IV based on the so-called "Swiss formula" that governed the Tokyo Round tariff liberalization to provide evidence of its effects on imports and inequality. Instrumented tariff reductions explain approximately 20% of the rise in income inequality between non-production and production workers between 1979 and 1988. This effect is largest among women, workers in routine occupations, and in more technology-intensive industries, suggesting a complementarity between trade liberalization and skill-biased technological change.

JEL codes: F13, F14, F66

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1 Introduction

Over the course of more than a decade, an enormous body of work has documented large and persistent consequences of trade for US firms and workers, exploiting differential exposure across geographic locations (Autor et al., 2013), industries (Pierce and Schott, 2016), or some combination of the two (Hakobyan and McLaren, 2016). In doing so, this literature has overwhelmingly focused on the post-1990 era of "hyper-globalization." This focus, however, comes with an important but often overlooked caveat: by the 1990s, much of the post-war US trade liberalization process was already complete. The US ad valorem equivalent (AVE) tariff, defined as the ratio of duties collected to total imports, stood at approximately 20% following the Tariff Act of 1930 – the so-called "Smoot-Hawley" tariff. By 1990, this had fallen to 3.3%.[1] This suggests that much of the economic impact of trade on the US economy was likely felt in the decades prior to those emphasized by scholars.

In particular, the distributional effects of the Toyko Round of the General Agreement on Tariffs and Trade (GATT) – a multilateral liberalization that governed US tariff reductions between 1980 and 1987 – have received almost no attention. This is despite the fact that, in terms of implied aggregate tariff changes, the Tokyo Round represents a larger liberalization than the Canada-US Free Trade Agreement (CUSFTA) in 1989, the North American Free Trade Agreement (NAFTA) in 1994, or the granting of Permanent Normal Trade Relations (PNTR) to China in 2001. Between 1979 and 1987, the average tariff across US manufacturing imports fell by 2.5 percentage points, resulting in a 1.6 percentage point decline in the US average tariff. The corresponding reductions in the US average tariff are 1.07 percentage points under PNTR, 0.75 percentage points under CUSFTA, and 0.14 percentage points under NAFTA.[2] Indeed, from this perspective, the Tokyo Round tariff cuts were larger than PNTR and more than twice the size of CUSFTA and NAFTA.[3]

[1] https://www.usitc.gov/documents/dataweb/ave_table_1891_2016.pdf

[2] The average tariff decline across manufacturing and non-manufacturing imports was 4 percentage points for Canada under CUSFTA (Trefler, 2004), which at its onset in 1989 applied to 18.8% of US imports ($-4 \times 0.188 = -0.75$ percentage points.) Similarly, a 2.1 percentage point cut for Mexico under NAFTA (Hakobyan and McLaren, 2016) impacted only 6.7% of imports in 1993. And while tariffs on Chinese imports did not change following PNTR, Handley and Limão (2017) estimate a reduction in uncertainty equivalent to a 13 percentage point reduction in tariffs, covering 8.2% of US imports in 2000 (USITC Dataweb). The manufacturing share of 1978 US imports was 63.5%, with the Tokyo Round liberalization covering more than 95% of all 1978 US imports and approximately 75% of these imports facing strictly positive tariffs.

[3] One potential reason that prior work has overlooked the Tokyo Round is that, as we detail in Greenland et al. (2025), the substantial tariff cuts in this period were accompanied by a compositional shift in US imports towards high tariff goods that left the import-weighted average tariff relatively constant.

In this paper, we explore the effects of the Tokyo Round liberalization on inequality between production and non-production workers in the US manufacturing sector. Our approach exploits newly available data on US tariffs in the era (Greenland et al., 2025) and a unique feature of the Tokyo Round negotiations that allows us to circumvent standard concerns regarding endogenous tariff policy. We instrument for US tariff changes between 1979 and 1988, exploiting the fact that the so-called "Swiss formula" served as the starting point for Tokyo Round negotiations. The Swiss formula compressed the tariff distribution by assigning larger tariff reductions to goods with higher pre-Tokyo tariff levels. Crucially, the only reason for differential liberalization across goods under the Swiss formula was differential pre-Tokyo tariffs.[4] Instrumenting for observed tariff changes with those dictated by the Swiss formula thus allows us to avoid the concern that US policymakers may have protected certain industries in this period due to, e.g., expected import growth. To address the concern that pre-Tokyo tariffs reflect historical protection choices of policymakers, our instrument replaces pre-Tokyo "column 1" (MFN) tariffs in the Swiss formula with their "column 2" counterpart. Column 2 tariffs were primarily determined by the Tariff Act of 1930 (Pierce and Schott (2016); Handley and Limão (2017)) and were reserved for a small subset of communist countries at the onset of the Tokyo Round. The exclusion restriction for our "Swiss IV" thus rests on any unobserved factors that drive changes in 1980s labor market outcomes not depending on 1930 tariffs.

We first use this instrument to estimate the impact of Tokyo Round tariff cuts on US industry-level trade flows between 1979 and 1988. Our estimated trade elasticity of 9 falls within the typical range for similar empirical settings (Ruhl, 2008; Head and Mayer, 2014). Moreover, we show that our results are robust to a wide range of controls for alternative trade policy shocks in the era. Ultimately, we argue that our Swiss IV has strong claims to reflecting exogenous tariff cuts.

The instrument reveals an important role for tariff changes in driving within-industry pay inequality between non-production and production workers.[5] We find that an interquartile increase in the magnitude of the industry-level tariff liberalization throughout our sample period increases log pay inequality by approximately 0.08. We show that import tariff liber-

[4]Our approach is part of a large body of work in which the extent of liberalization depends on pre-liberalization tariff levels. See work on, e.g., Chile (Pavcnik, 2002), Colombia (Goldberg and Pavcnik, 2005), India (Topalova, 2010; Topalova and Khandelwal, 2011), CUSFTA (Romalis, 2007; Kovak and Morrow, 2022), NAFTA (Caliendo and Parro, 2015; Hakobyan and McLaren, 2016; Besedes et al., 2020; Benguria, 2023), and the US granting normal trade relations (NTR) status to Vietnam (McCaig and Pavcnik, 2018) and PNTR to China (Pierce and Schott, 2016; Handley and Limão, 2017; Greenland et al., 2019).

[5]Throughout the paper, we refer to income and pay interchangeably.

alization can account for nearly 20% of the observed increase in log pay differentials between production and non-production workers in our sample period, with all of this occurring within as opposed to between industries. We decompose this effect into changes in the relative employment of non-production to production workers and changes in relative wages between worker types and show that the increase is primarily driven by the employment margin.

Exploring heterogeneity across workers, occupations, and industries, we document a striking pattern: the increase in pay inequality in response to the Tokyo Round is driven overwhelmingly by women. Indeed, the magnitude of the effect of falling tariffs on rising inequality among women is approximately double that of the manufacturing sector as a whole. We show that this is due in part to the fact that rising inequality within manufacturing is most pronounced in routine occupations, which account for a disproportionate share of women's employment in the era. Turning to the role of technological change, we find evidence that trade and technology interact to shape inequality in this period. In particular, we show that technology-intensive industries, as captured by higher pre-existing levels of automation, drive the increasing income inequality that accompanies the Tokyo Round liberalization.

While we see the paper as an important case study of the Tokyo Round, it also complements the large literature on 1980s trade and inequality more generally. Early work in this literature struggled to identify trade-driven changes in goods prices.[6] Furthermore, much of this literature focused on inequality between industries, often through the lens of factor endowment theory via the Heckscher-Ohlin model. In contrast, the broader labor literature documented that most changes in skill intensity happened within industries and especially in occupations and industries with more scope for adopting new skill-biased technologies (Acemoglu and Autor, 2011). Thus, the early literature concluded that trade had little effect on inequality. Our findings, conversely, reveal a crucial role for trade liberalization driving increased inequality within industries and especially in occupations and industries with more scope for adopting new skill-biased technologies.

More recently, Batistich and Bond (2023) identify notable effects on racial inequality in US labor markets due to rising imports from Japan in the 1970s and 1980s. In particular, they find relative reductions in manufacturing employment and labor force participation and

[6] See Slaughter (2000) and Feenstra and Hanson (2003) for excellent surveys. Slaughter (2000) in particular notes that this literature "made substantial progress understanding how to relate a given change in relative product prices to changes in relative factor prices ... [but] made less progress understanding whether these product-price changes have any thing to do with international trade."

declining wages for black males compared to white males. We differ from these prior studies by focusing on a broad-based, multilateral policy shock with an approach that does not rely on assumptions about the exogeneity of export growth in foreign markets, but rather on the institutional details of historical US tariff policy.[7] While complementary to existing work, our results emphasize an important role for trade policy in explaining inequality. The paper also contributes to the numerous studies exploring the relationship between trade and technology. Consistent with the disparate theoretical predictions on the link between import competition and innovation (Lawrence, 2000; Acemoglu, 2003; Aghion et al., 2005), empirical evidence on this relationship is mixed.[8] We underscore the fact trade and technology jointly, rather than separately, drive inequality.

A small number of papers have explored the effects of trade policy in this era. In Greenland et al. (2025), we use annual US legislated tariff data between 1972 and 1988 to show that the protection provided by specific tariffs was eroded by 1970s inflation despite the absence of tariff policy changes, and that this "accidental liberalization" was larger in magnitude than the Tokyo Round legislated tariff cuts. Most other work exploring the Tokyo Round took place during the early years of its phase-in and focused on predicted effects. Several papers emphasize consequences for the structure of production across member countries using general equilibrium models (Brown and Whalley, 1980; Deardorff and Stern, 1981, 1983), while others focus on the nature of the negotiations themselves (Ahmad, 1978; Chan, 1985). Closer to us, Gaston and Trefler (1994) explore the labor market consequences of tariffs in the period. Using cross-industry variation in 1983 US tariffs, they find that more protected industries paid higher wages but find no differential impact on imports. Moreover, they argue that policymakers did not manipulate the Tokyo Round tariffs in ways that affected labor market outcomes or imports. In contrast, our Swiss IV suggests a substantial role for endogenous policy setting.

The paper proceeds as follows. Section 2 describes the data. Section 3 describes our estimation strategy and identification challenges. Section 4 describes baseline results and robustness, while Section 5 explores heterogeneity in these responses. Section 6 concludes.

[7] In separate work, Lu and Ng (2013) instrument for US import penetration with UK import penetration and find that US manufacturing industries between 1971 and 2001 respond to import competition by substituting away from routine occupations towards non-routine occupations. This requires the assumption that UK import penetration is driven by supply-side forces in the rest of the world and not, for instance, UK demand shocks that are correlated with those in the US. Our approach requires no such assumption.

[8] For instance, Bustos (2011), Mion and Zhu (2013) and Bloom et al. (2016) find positive effects of trade on innovation, while Autor et al. (2020) and Aghion et al. (2024) find negative effects.

2 Data

To evaluate the effects of the Tokyo Round on inequality, we require information on industry outcomes, tariffs, and trade flows. Our final dataset draws from a number of sources. Primary industry outcomes come from the NBER-CES Manufacturing Database (Becker et al., 2021) and from IPUMS (Ruggles et al., 2024). From these datasets we collect information on wages and employment at the industry level separately for production and non-production workers.[9] IPUMS data also allow us to separate workers by sex and occupation. The NBER data are available annually, while IPUMS data are reported for each census year. Annual industry imports and exports come from Schott (2008). The NBER and trade flow data measure industry outcomes at the 4-digit level of the 1987 Standard Industrial Classification (SIC) system, while IPUMS data are available at the Census industry level. Finally, we take tariff line-level legislated US tariffs from Greenland et al. (2025) and, as Appendix B.1 describes in detail, concord them to calculate AVE tariffs at the industry level.[10] Appendix Table A.1 contains summary statistics.

Figure 1 about here.

Before moving to our empirical methodology in Section 3, we briefly describe the trade environment as well as the evolution of inequality during our sample. To this end, Figure 1 presents a heat map describing 1980s US import growth across sectors and partner countries. Descending along the vertical axis are the largest exporters to the US in 1978, with Canada and Japan comprising more than 40% of US imports immediately preceding the Tokyo Round. The horizontal axis orders 2-digit SIC sectors from largest to smallest in terms of the sector's 1978 share of US imports. Cooler colored squares in the figure represent smaller log import growth between 1979 and 1988, while warmer colors represent larger growth.

The key takeaway from Figure 1 is that, unlike many other episodes studied in the literature on trade and labor in the US, import growth following the Tokyo Round is not limited to a few countries nor to a few sectors. Illustrating the breadth of import growth across countries, log imports from every country except Germany grow by at least 1.25 (250%) in at least two sectors. Similarly, log imports in every sector except agriculture see

[9] We use "non-production" and "production" to be consistent with the definitions in the NBER-CES data. Scholars from the era alternatively referred to these groups as "skilled" and "unskilled", respectively.

[10] Because the US uses both ad valorem and specific tariffs in this era, the dataset contains legislated ad valorem tariffs and the AVE of legislated specific tariffs that together sum to the AVE legislated tariff for the good. The AVE of a legislated specific tariff is the legislated specific tariff divided by its unit value. The dataset covers over 5000 5-digit TSUSA goods and over 98% of imports during our sample.

growth of at least 1.25 from at least two countries. Moreover, import growth is negative only for Japanese primary metals manufacturing and Hong Kong petroleum. That is, while import growth is most pronounced in high-tech sectors such as electronics equipment, computer equipment, and measuring equipment and from countries that were not the top few US trade partners – i.e., countries other than Canada, Japan, and Germany – import growth is not limited to particular trade partners or industries.

Figure 2 about here.

This broad-based growth in imports mirrors the nature of the tariff liberalization following the Tokyo Round. Figure 2 displays the 4-digit SIC AVE tariffs in 1978, with each dot representing one of the 395 industries, and with industries grouped according to their 2-digit SIC sector. On average, 4-digit industries have a pre-Tokyo AVE tariff of 7.1%. However, the variation is considerable, both within and across sectors, with a standard deviation of 6.2%. The variation in pre-Tokyo AVE tariffs corresponds directly to variation in Tokyo Round tariff cuts since, as we discuss in detail below, the Swiss formula reduces tariffs by more for industries with higher pre-Tokyo AVE tariffs. In sum, the Tokyo Round led to large and varying exposure to tariff liberalization both within and across sectors.

Figure 3 about here.

Finally, we display the industry-level variation in changes in inequality throughout our sample. Figure 3 displays three kernel densities of log changes between 1979 and 1988. The gray distribution outlined in black represents changes in income paid to non-production relative to production workers. The blue and red distributions represent the two components of income inequality: employment of non-production relative to production workers and the wage of non-production relative to production workers, respectively. Two points stand out in the figure. First, more than 80% of industries see rising income inequality, producing an average increase of approximately 0.17 log points. Second, this is accompanied by rising non-production employment intensity which grows, on average, by 0.16 log points. In contrast, on average, wage inequality increases by a modest 0.01 log points, with a distribution that is nearly symmetric around 0. In formally exploring the effects of the Tokyo Round liberalization along each of these margins, we find a similarly dominant role for the employment channel in driving income inequality. We return to this below.

3 Empirical Methodology

We explore the effects of tariff liberalization under the Tokyo Round on US trade flows and industry outcomes. The sample period for our outcome variables in the trade and NBER-CES manufacturing data begins in 1979 – the year before the US begins implementing Tokyo Round tariff cuts – and ends in 1988 – the year after the US completes implementing the liberalization. In our IPUMS specifications, we examine changes between 1980 and 1990. In both settings, we estimate industry-level first-difference regressions of the form

$$\Delta y_i = \alpha + \beta_0 \Delta \ln(1 + AVE_i) + \boldsymbol{\beta} \mathbf{X}_i + \varepsilon_i \tag{1}$$

where, for industry i, Δy_i is the change in a trade flow or other outcome of interest between the first and last year in the sample and $\Delta \ln(1 + AVE_i)$ is the log change in one plus the industry's AVE tariff. $\mathbf{X}_i$ is a vector of industry-specific controls and ε_i is the error term.

Equation (1) faces the standard identification concern that tariffs are set non-randomly. The specific concern in our context is that industry-level tariff reductions may be correlated with industry-level trends in import growth or labor market outcomes. For example, policymakers may choose to protect industries facing high expected import growth by assigning them smaller tariff reductions. Such endogenous policy setting would bias a negative coefficient of tariff changes on imports upwards towards zero, as larger tariff cuts would be associated with lower average import growth. Alternatively, against a backdrop of rising inequality and surging import competition, policymakers may choose to protect industries in which they expect rising inequality to be most pronounced. This would again bias any negative estimate of the relationship between tariff changes and changing income inequality upwards towards zero, as larger tariff cuts would be associated with lower growth in inequality.

To address these concerns, we rely on a unique institutional detail of the Tokyo Round negotiations. In part due to fears that strategic protectionism would hamper the liberalization process, parties to the Tokyo Round sought a formula-based approach similar to the one taken under the Kennedy Round, which had aimed to reduce tariffs by 50%. While the US proposed a 60% reduction on all industrial tariffs under the Tokyo Round, the EEC preferred "harmonization", in which initially higher tariffs were reduced by more (Neu, 1979). Consistent with this aim, the Swiss delegation to the negotiations proposed a simple formula for post-Tokyo tariffs that depended only on a good's initial tariff. This so-called "Swiss

formula" compressed the distribution of final AVE tariffs in year T by imposing a maximum AVE tariff of 0.14 and assigning larger tariff cuts to products with larger tariffs in an initial year t_0 (Swiss Delegation, 1976).[11] Specifically, the Swiss formula is

$$AVE_{iT}^{Swiss} = \frac{0.14 \times AVE_{it_0}}{0.14 + AVE_{it_0}}. \tag{2}$$

The extent of an industry's liberalization under the formula is thus completely determined by initial AVE_{it_0} tariffs, which are the only potential source of endogeneity in the tariff liberalization implied by the formula.[12] Importantly for our identification, Neu (1979) notes that "there was no historical or intellectual reason for the choice of this particular formula, it was chosen principally because it was simple and implied an acceptable average depth of cut" (p. 11). This suggests that the formula itself was not the product of political economy considerations related to, e.g., US imports and labor market outcomes.

Figure 4 about here.

That said, the extent that countries followed the Swiss formula varied both across countries and across sectors within a country. While the US followed the Swiss formula much more than other countries, they did not strictly adhere to the Swiss formula (see, e.g. Table 14 of Deardorff and Stern (1979) or Table 6 of Neu (1979)). The US identified sectors including textiles, apparel, leather, and footwear as needing protection and they indeed faced less than Swiss-implied tariff cuts. Other US sectors such as crops, tobacco and chemicals faced less than Swiss-implied tariff cuts likely due to other countries imposing less than Swiss implied tariff cuts (Neu, 1979).

We explore the ultimate results of this process in Figure 4, which plots the 4-digit SIC AVE tariff change between 1979 and 1988 implied by the Swiss formula on the horizontal axis against the observed AVE tariff change over this period on the vertical axis.[13] The grey 45 degree line represents exact adherence to the Swiss formula, such that industries above the line are "under-liberalized" and receive less-than-prescribed tariff cuts while industries below the line are "over-liberalized" and receive more-than-prescribed cuts. Three points

[11] The tariff cut implied by the Swiss formula tariff is $AVE_{iT}^{Swiss} - AVE_{it_0} = \frac{-AVE_{it_0}^2}{0.14+AVE_{it_0}}$, which is increasing in AVE_{it_0}. Further, the maximum final tariff is $\lim_{AVE_{it_0}\to\infty} \frac{0.14\times AVE_{it_0}}{0.14+AVE_{it_0}} = \lim_{AVE_{it_0}\to\infty} \frac{0.14}{\frac{0.14}{AVE_{it_0}}+1} = 0.14$.

[12] The Swiss delegation proposed the value 0.14 and, in practice, the US followed the proposal and used this value (Swiss Delegation, 1976; Deardorff and Stern, 1979).

[13] See Figure A.1 in Appendix **??** for an analogous version of Figure 4 at the 5-digit TSUSA tariff line level.

stand out in the figure. First, as discussed above, Tokyo Round tariff changes constitute a substantial liberalization: the mean observed tariff reduction is 2.5 percentage points, with a standard deviation of 2.7 percentage points. Second, the strong positive correlation of 0.64 between observed and prescribed tariff cuts implies that the Swiss formula does indeed serve as a benchmark for tariff liberalization under the Tokyo Round. Finally, the dispersion across industries in observed tariff cuts for a *given* Swiss-prescribed tariff cut – that is, the extent of the deviation from the 45 degree line – is an empirical representation of the concerns regarding endogenous tariff setting that motivate our instrument. In Appendix C, we explore the extent of under- or over-liberalization in more detail and show that industries with greater production shares of labor tended to be under-liberalized.

While tariff cuts under the Swiss formula are determined entirely by initial tariffs and a single known parameter, these initial tariffs are themselves non-random. In particular, as we show in Appendix C, industries receiving more protection on the eve of the Tokyo Round use production labor more intensively. As such, using the tariff variation implied by the Swiss formula may still generate biased estimates through the endogenous relationship between outcomes and initial tariff levels.

To address this concern, we rely on a second institutional feature of the US tariff schedule: while the US imposes so-called "column 1" (i.e. MFN) tariffs on imports from most countries, it imposed "column 2" tariffs on imports from a small subset of communist countries at the onset of the Tokyo Round.[14] These column 2 tariffs were themselves originally the column 1 tariffs from the Smoot-Hawley Tariff Act of 1930.[15] This implies that the column 1 tariffs immediately preceding the Tokyo Round reflect the six GATT rounds between 1930 and 1979 and, in turn, the decisions of policymakers to protect certain industries during these rounds. However, they also reflect the 1930s column 1 tariffs themselves – that is, Tokyo-era column 2 tariffs. As a result, there is a strong positive correlation between the pre-Tokyo column 2 tariffs and their column 1 counterparts. But, as they were set nearly 50 years prior to our period of study, the column 2 tariffs are less likely to have been set with an eye

[14] As is well known, GATT Article I requires that GATT members apply MFN tariffs on imports from all member countries. This, however, leaves discretion for the tariffs imposed on non-members. The extent of discretion is substantial at the onset of the Tokyo Round, when fewer than 90 countries belonged to the GATT. In practice, during our sample period the US imposes column 1 tariffs on the vast majority of non-GATT members and imposes column 2 tariffs on a small subset of communist non-GATT members, but violates GATT Article I by imposing column 2 tariffs on some communist GATT members.

[15] Products have, of course, been added to the tariff schedule since 1930. New products are typically classified by the Committee for Statistical Annotation of Tariff Schedules, more commonly known as the 484(f) Committee, by assigning them into existing tariff categories, including general "other" classifications. (https://www.usitc.gov/harmonized_tariff_information/modifications_to_hts).

towards 1980s outcomes of interest than the pre-Tokyo column 1 tariffs.[16]

The construction of our instrument rests on each of the two institutional features of US tariff policy described above: that Tokyo Round negotiations used the Swiss formula as a benchmark and that column 2 tariffs were determined by the Tariff Act of 1930 but were still strongly correlated with their column 1 counterparts on the eve of the Tokyo Round. Specifically, we rely on the Swiss formula to dictate the magnitude of tariff cuts across goods g but substitute column 2 AVE tariffs into the formula in lieu of the observed column 1 AVE tariffs. At the 5-digit TSUSA level, we construct time-varying values of the Swiss measure. The value in 1988, after the phase-out has been completed, is given by

$$AVE^{Swiss}_{g,1988} = \frac{0.14 \times AVE^{Col2}_{g,1978}}{0.14 + AVE^{Col2}_{g,1978}}. \tag{3}$$

The value in 1979, prior to the liberalization, is the 1978 column 2 AVE tariff, $AVE^{Col2}_{g,1978}$.[17] We aggregate each of these to the 4-digit SIC level, as detailed in Appendix B.1. Finally, we construct our instrument as the change in the log of one plus the constructed Swiss measure:

$$\Delta \ln(1 + AVE^{IV}_i) = \ln(1 + AVE^{Swiss}_{i,1988}) - \ln(1 + AVE^{Col2}_{i,1978}) \tag{4}$$

Figure 5 illustrates the two-stage least squares intuition behind our Swiss IV at the 4-digit SIC level. The left panel plots the first stage relationship: the instrumented change in AVE tariffs between 1979 and 1988 on the horizontal axis against the observed change in tariffs over this period on the vertical axis.[18] Our instrumented change in AVE tariffs strongly predicts observed tariff changes: industries with higher 1978 column 2 AVE tariffs, and thus greater predicted cuts by the Swiss formula, see substantially larger AVE tariff cuts following the Tokyo Round.

Figure 5 about here.

[16] Appendix Figure A.2 illustrates the strong positive correlation between column 1 and column 2 tariffs at the tariff-line TSUSA level. It also shows substantial dispersion across column 1 tariffs for a given column 2 tariff, implying that policymakers did indeed protect some industries more than others during the six preceding GATT rounds. Figure A.3 produces the same figure for 4-digit SIC industries in 1978.

[17] We construct the instrument using the AVE in 1978 instead of 1979 to avoid concerns regarding endogenous determination of contemporaneous unit values. Our results are unchanged by using 1979 AVE tariffs. In exercises exploring shorter horizons than the full 1979-1988 period, we allow the measure AVE^{Swiss}_{gt} to phase out linearly from $AVE^{Col2}_{g,1978}$ to AVE^{IV}_{gT}, following the approach taken in practice.

[18] A small number of industries see AVE tariff increases. This is possible even in the absence of legislated tariff increases due to falling unit values in the presence of specific tariffs. Approximately one-third of US tariffs lines before the Tokyo Round faced specific tariffs (Greenland et al., 2025).

The right panel of the figure displays the second stage relationship, with the predicted AVE tariff change from the left panel on the horizontal axis plotted against observed import growth on the vertical axis. The figure suggests that the Tokyo Round liberalization generated substantial import growth, as industries with larger predicted tariff reductions exhibit stronger import growth between 1979 and 1988. This relationship also suggests a role for Tokyo Round tariff cuts in explaining other industry outcomes in this era.

The key exclusion restriction for our instrument is that any unobserved factors determining the Swiss-implied tariff liberalization do not also directly affect 1980s US trade or labor market outcomes.[19] As the cross-industry variation in the measure stems entirely from variation in column 2 tariffs, this amounts to a requirement that correlates of 1930 tariffs do not determine 1980s changes in inequality. Despite being set 50 years prior to our sample period, the Smoot-Hawley tariffs were themselves the outcome of a political process in which firms' and workers' lobbying influence played a critical role (Irwin and Kroszner, 1996; Irwin and Soderbery, 2021). To the extent that the determinants of lobbying influence in 1930 also affect 1980s labor markets, our results may be biased. To explore this, we conduct a number of balance tests. In particular, we regress $\Delta \ln(1 + AVE_i^{IV})$ against three categories of industry-level characteristics: other trade-related measures, measures of industry factor intensity, and measures of exposure to skill-biased technological change.

Our trade-related measures are as follows. First, to account for changes in export opportunities as a result of the Tokyo Round liberalization, we include an import-weighted average of changes in tariff rates between 1979 and 1987 in Canada, Japan, and the European Economic Community (EEC).[20] Second, we include log changes in industry unit values between 1972 and 1979 and the share of the tariff-inclusive price in 1979 accounted for by specific tariffs.[21] These jointly capture exposure to inflation-driven changes in industry-level protection in the presence of specific tariffs (Greenland et al., 2025). Finally, we include a

[19] See Appendix B.2 for a formal derivation of our exclusion restriction.

[20] For Japan and the EEC, we digitize tariff schedules from the WTO website: `https://www.wto.org/english/docs_e/gattbilaterals_e/indexbyround_e.htm`. We manually concord tariff lines to the Customs Co-operation Council Nomenclature (CCCN) and map these to 1992 HS codes using a concordance provided by WITS: `https://wits.worldbank.org/product_concordance.html`. Finally, we concord these to the SIC using the concordance created by Pierce and Schott (2012). We take Canadian import tariffs from the Canadian SIC classification and concord these to the US SIC classification using data and a concordance from Trefler (2004). We chose these countries due to their importance as US export destinations as displayed in figure B.1. A scatterplot of industry import tariff changes and export tariff changes may also be found in figure B.2.

[21] This can be written as $STS_i \frac{AVE_i}{1+AVE_i}$, where STS_i is the share of tariff revenue in industry i generated by specific tariffs.

dummy variable equal to 1 if the industry was covered by the Multifiber Arrangement, which imposed import quotas on textiles and apparel (Harrigan and Barrows, 2009; Khandelwal et al., 2013).

For industry factor intensity, we include log capital-to-labor ratios, the share of employment accounted for by non-production workers, and the ratio of materials to shipments as of 1978, as well as the share of employment accounted for by women as of 1980.[22] For skill-biased technological change, we include log 1978 investment and changes in log investment between 1970 and 1978, both using data from the NBER-CES data. We also include the routine task-intensity index of Autor and Dorn (2013), which is calculated based on data from the Dictionary of Occupational Titles as of 1977, and changes in automation between 1947 and 1978, following Acemoglu and Restrepo (2019, 2020).[23]

Table 1 about here.

In Table 1, we regress our instrument against alternative trade measures in column 1, factor intensity measures in column 2, technological change measures in column 3, and the three simultaneously in column 4. As is clear from the table, our Swiss IV is indeed correlated with a number of industry characteristics.[24] The measure is positively correlated with industry-level tariff changes in trade partners, suggesting that industry-level liberalization choices were correlated across countries. We also find a positive correlation with unit value changes between 1972 and 1979 and a negative correlation with MFA coverage. Tariff changes implied by our instrument are positively correlated with 1978 log capital-labor ratios and non-production labor shares, suggesting greater liberalization in industries intensive in production labor. This is consistent with protection for labor-intensive industries in the Smoot-Hawley era. We also find a negative correlation with women's employment shares. As

[22] We calculate capital-to-labor ratios, non-production employment shares, and materials to shipments with data from the NBER-CES database. Data on women's employment shares are calculated based on data from IPUMS, which we concord from the Census industry to the SIC classification.

[23] The routineness measure of Autor and Dorn (2013) is defined at the occupation level - specifically, at the *occ1990dd* classification – as the log of routine task inputs minus the log of manual task inputs and the log of abstract task inputs. We concord this to the Census industry classification using weights based on 1980 IPUMS data. We then concord to the SIC level using a concordance from Lake and Millimet (2016). We take data on automation – specifically "displacement", or reductions in labor's share of value added after controlling for factor price movements – at the Census industry classification between 1947 and 1978 from Acemoglu and Restrepo (2019) and Acemoglu and Restrepo (2020) and concord this to the SIC level using concordances from Autor et al. (2019).

[24] As noted above, and formally derived in Appendix B.2, our exclusion restriction revolves around the column 2 tariffs. To this end, we note that the qualitative results from a version of Table 1 with the column 2 tariff as the dependent variable remain the same (although, naturally, the point estimates differ). More generally, the issue of whether the column 2 tariffs are correlated with modern industry-level characteristics is something the literature has largely ignored.

with the MFA correlation, this is consistent with the fact industries such as textiles and apparel faced large predicted tariff reductions. Finally, the instrument is positively correlated with material shares of shipments and automation and negatively correlated with industry routineness and 1970s investment growth. Overall, larger predicted tariff reductions correlate with lower investment levels and automation and more routine occupations.

To the extent that the characteristics correlated with our instrument also drive inequality in the 1980s, our estimates will be biased. We take several steps to address this concern. First, we include the measures from Table 1 in our baseline estimates and show that our results are robust to their inclusion. Second, in robustness exercises, we control for lagged changes in our outcomes of interest. To the extent that omitted variables drive our results even after conditioning on the industry-level controls described above, controlling for lagged changes in outcomes eliminates the persistent relationship between them and the omitted variables. Our results, again, are largely unaffected by this change. Finally, we conduct placebo analyses, exploring the relationship between changes in inequality between 1972 and 1979 and the Swiss IV. If anything, we find the opposite relationship from the one observed in the 1980s – that is, inequality was falling in the 1970s in the industries that were ultimately more exposed to the Tokyo Round, and this trend was reversed with the implementation of the tariffs. We discuss our results, including these robustness exercises, in detail now.

4 Results

4.1 Trade flows

To the extent that tariffs affect labor market outcomes, they do so through their effect on goods prices and, in turn, trade flows. With this in mind, in Table 2 we explore the relationship between industry-level tariff liberalization and trade flows. Column 1 presents the effect of industry tariff changes on log import growth using an OLS specification.[25] While the point estimate has the expected sign, the magnitude is small, with an elasticity near 1.7, and the effect is not statistically significant. In column 2, we instrument for observed AVE tariff changes with our Swiss measure. Underscoring the importance of addressing endogeneity in tariff changes in this era, the results in the column change substantially.

[25]To maintain a consistent sample size for each column, we restrict our sample to the 392 industries for which we are able to calculate all outcomes. Summary statistics for these outcomes may be found in Appendix Table A.1.

The relationship remains negative, but the elasticity rises to 9 and is significant at the 1% level. This elasticity, while large, is consistent with other analyses of trade liberalizations.[26] Further, consistent with the discussion in Section 3, the first-stage F-statistic suggests that the instrument is strong. We also note that the bias upwards toward zero evident in column 1 places structure of the nature of the endogeneity concern. Specifically, it suggests that policymakers protected industries in which they expected more rapid import growth.

Table 2 about here.

In columns 3 and 4 we examine the robustness of this result to controlling for other trade policy measures, as discussed in Table 1 above. A notable feature of the US tariff schedule in this era is the fact that many tariffs were specified in per-unit, or *specific* terms. Indeed, as of 1979, approximately 17% of tariff revenue in the manufacturing sector was generated by specific rather than ad valorem tariffs and roughly 70% of the industries in our sample have specific tariffs on some products. This distinction is important, as the protection afforded by specific tariffs varies inversely with the price level (Greenland et al., 2025).[27] This implies that industries may be differentially exposed to 1980s price variation as a function of dependence on specific tariffs. Additionally, in the presence of specific tariffs, 1970s price changes directly affect pre-Tokyo AVE levels.

To address these issues, in column 3 we control for the change in log import unit values between 1972 and 1979 and for the 1979 industry-level share of tariff-inclusive prices accounted for by specific tariffs. Inclusion of these controls increases the estimated elasticity slightly, to 9.7, but leaves the results qualitatively unchanged. In column 4 we control for more explicit trade policies of the era. First, we introduce a dummy variable equal to 1 for industries with products subject to MFA restrictions. As discussed above, textiles and apparel saw large tariff reductions under the Tokyo Round.[28] For these industries, it is important to separate tariffs from other non-tariff barriers. Second, we account for the fact that, while US import tariffs declined under the Tokyo Round, so too did the tariffs on US exports to GATT member countries. As the extent of industry-level liberalization was correlated across countries, it is possible that the estimated elasticity captures some component of the effects of tariff liberalization on US exports. We address this possibility by controlling for the extent of tariff liberalization in Canada, Japan, and the European Economic Community (EEC).

[26] See, e.g., Ruhl (2008) and Head and Mayer (2014).

[27] For detailed discussions of the importance of specific tariffs in historical contexts, see Crucini (1994), Irwin (1998), and Greenland and Lopresti (2024).

[28] Textiles and apparel were exempted from the full Swiss-implied cuts, but still saw larger than average tariff reductions.

As can be seen in column 4 of the table, controlling for the MFA and tariff liberalization on US exports reduces the point estimate of interest slightly, but leaves it large and statistically significant.

It is possible that consumption responds to the Tokyo Round liberalization, and imports respond mechanically as a result. To explore this possibility, in column 5 we repeat the specification from column 4 with changes in import penetration, defined as imports relative to output consumed domestically – i.e., shipments less net exports – as our outcome of interest. We find similar results, with the Tokyo Round liberalization increasing import penetration in more exposed industries relative to less exposed ones. To compare magnitudes across columns 4 and 5, note than an interquartile increase in the extent of industry-level tariff liberalization reduces log import growth by approximately 37% of its interquartile range, while the analogous value for import penetration is slightly larger at 54%. Tariff liberalization plays a substantial role in explaining changes in US imports, however defined, throughout the 1980s.

While our focus is the effect of the Tokyo Round on imports, it may be the case that exports respond as well, either through indirect effects of tariff liberalization on intermediate inputs or due to the multilateral nature of the agreement. To explore this possibility, columns 6 and 7 examine the impact of tariff changes on growth in log exports and exports relative to shipments, respectively. While the point estimate of interest in both columns is negative, neither is statistically significant. Further, the export elasticity in column 6 is less than half of the import elasticity in column 4. Taken together, this suggests that industry-levels exports respond weakly, if at all, to Tokyo Round import tariff cuts.

Ultimately, then, Table 2 delivers two key messages. First, Tokyo Round import tariff reductions in the US have large effects on imports but no statistically significant effect on exports.[29] Second, failing to account for endogenous tariff policy in the Tokyo era yields substantially biased results. In particular, policymakers appear to have protected industries that were more exposed to import growth during the liberalization, highlighting the value of our instrument in isolating random variation in tariff cuts.[30]

[29] In Table A.2 we conduct a placebo exercise by regressing trade-related outcomes from 1972-1979 on our instrument. We find that the instrumented tariff changes are not predictive of import growth in the pre-Tokyo era.

[30] There is precedent for using initial tariff levels as an instrument for the magnitude of the tariff cut – see, e.g., Goldberg and Pavcnik (2005). In appendix Table A.3, we report estimates of import elasticities using the implied initial column 2 tariff as an IV instead. Given the structure of the Swiss formula, these predicted tariff changes are very highly correlated in logs and our results are essentially unchanged by this alternative approach (this applies for our subsequent inequality results as well). This highlights the value of

4.2 Within-industry pay inequality

We now turn to the impact of tariff liberalization on within-industry inequality, beginning with the NBER-CES Manufacturing database. Specifically, in Table 3 we explore how tariff changes affect industry-level log pay differences between non-production and production workers between 1979 and 1988.

Table 3 about here.

In column 1 of the table we regress changes in log pay differentials against changes in industry level $\Delta \ln(1 + AVE_i)$, instrumenting with our Swiss measure as above. The column shows that tariff liberalization leads to a relative increase in within-industry pay inequality, shifting payments towards non-production workers. In particular, an interquartile reduction in tariff levels increases the log pay differential by approximately 0.05 log points.[31]

In column 2, we introduce controls for trade policy, as discussed above. In particular, we control for changes in industry log unit values between 1972 and 1979, the share of the tariff-inclusive price accounted for by specific tariffs in 1979, whether an industry was exposed to the MFA, and the 1979-1988 change in AVE on industry exports. Inclusion of these controls increases the magnitude of the point estimate of interest considerably. The results in the column imply that an interquartile increase in industry tariff liberalization increases log pay differentials by approximately 0.09. Exposure to the MFA is particularly relevant. The point estimate on the control itself is negative and statistically significant, implying that industries covered by the MFA experienced relative reductions in log pay differentials between 1979 and 1988. Further, that its inclusion increases the point estimate on $\Delta \ln(1 + AVE_{i,t})$ is consistent with the fact that textiles and apparel experienced large reductions in tariffs under the Tokyo Round. While the tariff liberalization itself increases pay differentials, MFA coverage reduces them. Failing to control for MFA exposure thus biases the point estimate on tariff liberalization upwards toward zero.

In column 3 we introduce controls for industry factor intensity, including log capital-to-labor ratios, the share of employment accounted for by non-production workers, the share of employment accounted for by women, and material inputs relative to shipments, all as of 1978. Including these controls leaves our result essentially unchanged and statistically significant at the 1% level. This suggests that, while Table 1 reveals that these measures are in fact correlated with our instrument, they do not separately determine inequality in the

using the Swiss formula for understanding endogenous policymaking as we do in Appendix C.

[31] The interquartile range in observed AVE changes between 1979 and 1988 is equal to 0.029

1980s.

Finally, in column 4 we control for proxies of technological change: log industry investment as of 1978, log investment growth between 1970 and 1978, the routineness index of Autor and Dorn (2013), and a measure of automation between 1947 and 1978, following Acemoglu and Restrepo (2019, 2020). Automation, in particular, plays a role in explaining rising pay differentials in our data. However, the magnitude of our point estimate of interest is unaffected: an interquartile increase in tariff liberalization increases the skill pay differential by 0.08 log points.

Table 4 about here.

A primary concern for our approach is that changes in skill pay differentials in the 1980s may be a continuation of pre-existing trends that are coincident with but not driven by the Tokyo Round liberalization. While the industry controls in Table 3 address this to some extent, we explore the possibility more directly in Table 4. First, in Panel A we estimate the relationship between changes in pay inequality between 1972 and 1979 and the instrumented change in tariffs between 1979 and 1988, repeating the specifications from Table 3.[32] To the extent that our baseline results capture pre-existing labor market trends, we would expect similar results to those in the previous table. As is clear from the table, this is not the case. The coefficient of interest is insignificant in columns 1 and 2 and only significant at the 10% level in column 4. Moreover, the sign of the coefficient is reversed, suggesting that inequality was *falling* throughout the 1970s in industries that would ultimately be exposed to the Tokyo Round tariff cuts.

In Panel B, we directly account for pre-existing trends by re-estimating each column from Table 3 while controlling for 1972-1979 changes in log pay inequality. As is clear from the panel, this has little effect on our baseline results. Our preferred estimate, in column 4, falls in magnitude from -2.7 to -2.5, but remains significant at the 5% level. This, again, suggests that our results are not the result of pre-existing trends.[33]

The results above focus on the effect of changes in own-industry tariffs, but ignore any indirect effects through upstream liberalization. To explore the role played by changes in input tariffs in our findings, we create a weighted average of observed tariff changes on an

[32] We choose this period to align with our import regression, for which data are only available beginning in 1972. To facilitate comparison we scale up the 1972-1979 change to its 9-year equivalent change (i.e. multiplying by 9/7).

[33] In appendix Table A.4 we show an analogous result for our trade outcomes: Tariff changes are not related to pre-Tokyo trade outcomes, and their relationship with post-Tokyo outcomes isn't impacted by inclusion of lagged outcomes.

industry's upstream inputs. An input tariff receives greater weight in the average when the industry relies more on the input for its own production, as calculated using the 1977 BEA input-output table, and the input represents a larger share of 1978 US imports. To account for endogeneity of tariff changes on inputs, we also calculate this measure using our Swiss IV to instrument for upstream tariffs.[34]

Table 5 about here.

Table 5 repeats the specifications from Table 3 while additionally controlling for instrumented upstream tariffs.[35] Two points emerge from the table. First, the inclusion of upstream tariffs has little effect on our baseline estimates. In column 4, which includes the full set of baseline controls, the addition of $\Delta \ln \left(1 + AVE_{i,t}^{Up}\right)$ reduces the point estimate of interest slightly, but leaves our result qualitatively unchanged. Second, the effect of upstream tariff liberalization reinforces the effect of own-industry liberalization, albeit somewhat noisily. The estimates in column 4 imply that an interquartile reduction in upstream tariffs increases within-industry pay inequality by approximately 0.04 log points, though the effect is only statistically significant at the 10% level.

4.3 Alternative Time Horizons and Employment v. Wage Margins

Our baseline estimates examine changes between 1979 and 1988, spanning the year immediately preceding the implementation of the Tokyo Round liberalization to the year after the phase-in had been completed. In Panel A of Figure 6, we explore the relationship between tariff changes and pay inequality over different time horizons. In particular, we repeat the specification from column 4 of Table 3 separately for one- to nine-year changes between 1979 and the respective end years. For each specification, we allow the Swiss measure to phase-in linearly between 1979 and 1987, such that the change in the measure for the one-year change specification is equal to one-eighth of the change over the entire period, and so forth.

Figure 6 about here.

As is clear from panel (a) of the figure, our results are broadly robust to alternative time horizons. The estimated effects are largest over changes of one to four years, peaking at -6.7 in the three-year change specification and falling to -2.7 by the end of the sample. This suggests that labor markets may have adjusted over time in response to import growth, though the effect is large and statistically significant throughout the entirety of the sample

[34] We detail construction of these variables in Appendix B.3.

[35] We also estimate the effects of changes in upstream tariffs on imports in appendix Table A.5.

period.[36]

Rising income inequality within industries can reflect either rising wage inequality or a shift in the composition of labor towards non-production workers. We explore this distinction in panels (b) and (c) of Figure 6, decomposing income inequality into its employment margin, defined as the log ratio of non-production to production employment, and its wage inequality margin, defined as the log ratio of non-production labor payroll per non-production worker relative to production labor payroll per production worker. The figures reveal a dominant role for the employment margin. While somewhat less precise than the estimates for pay inequality, the employment estimates are negative at all time horizons save for one-year changes and are of similar magnitude to the pay inequality estimates. Point estimates on wage inequality, on the other hand, are close to zero throughout the sample and only become statistically significant at the 10% level in 1988.[37]

4.4 Inequality Within and Between Industries

Our analysis thus far emphasizes the relationship between tariff changes and pay inequality *within* industries. This, however, ignores any role for inequality *between* industries, such as might arise if exported industries expand and imported industries contract. To this end, we now use the within-between decomposition of Berman et al. (1994) to quantify the relevance of our within-industry analysis for observed changes in aggregate inequality.

Specifically, we decompose the observed change in the non-production share of US income between 1979 and 1988 into its within- and between-industry components. To that end, let $y_{NP} = \frac{Pay_{NP}}{Pay}$ denote the non-production share of US manufacturing income, $y_i = \frac{Pay_i}{Pay}$ denote industry i's share of manufacturing income, and $y_{NP,i} = \frac{Pay_{NP,i}}{Pay_i}$ denote the non-production share of income in industry i. Changes in aggregate inequality can then be decomposed as follows:

$$\Delta y_{NP} = \sum_i \bar{y}_i \Delta y_{NP,i} + \sum_i \bar{y}_{NP,i} \Delta y_i, \tag{5}$$

where $\bar{y}$ denotes the mean of variable y across the initial year 1979 and the end year 1988. The decomposition illustrates that the change in the aggregate non-production share of income depends on two components. First, the "within component" aggregates within-industry

[36] An analogous figure displaying the relationship for imports, import penetration, and exports relative to shipments may be found in Appendix Figure A.4.

[37] The table reporting this decomposition for nine-year changes analogous to Table 3 in which controls are introduced sequentially can be found in Appendix Table A.6.

changes in non-production income shares, $\Delta y_{NP,i}$, using an industry's mean share of income, $\bar{y}_i$, as the weight. Second, the "between" component aggregates changes in the between-industry shares of aggregate income, Δy_i, using industry-specific mean non-production shares of income, $y_{NP,i}$, as weights.

The average between-industry share of aggregate income, $\bar{y}_i$, and the average within-industry non-production share of income, $\bar{y}_{NP,i}$, each depend on their initial value in 1979 and the associated change between 1979 and 1988. Specifically, we can write

$$\Delta y_{NP} = \sum_i \left(y_{i,1979} + \frac{1}{2}\Delta y_{i,1979} \right) \Delta y_{NP,i} + \sum_i \left(y_{NP,i,1979} + \frac{1}{2}\Delta y_{NP,i,1979} \right) \Delta y_i. \quad (6)$$

Following equation (6), we can calculate the within and between components of the change in the aggregate non-production share of income stemming from the liberalization. This calculation requires four pieces of information: 1979 industry-level income shares, 1979 industry-level non-production income shares, and estimates of both the change in industry-specific employment shares and the change in industry-level non-production income shares caused by the Tokyo Round. We can then assess the aggregate importance of the Tokyo Round for US inequality by comparing the observed change in the US non-production income share to the change due to the Tokyo Round.

Figure 7 about here.

For the sake of space, we provide a full derivation and detail the estimation procedure in Appendix D. We report our results in Figure 7, which displays the share of aggregate changes in pay inequality accounted for by US tariff changes at time horizons between one and nine years.[38]. As is clear from the figure, the Tokyo Round increases inequality meaningfully, accounting for between 15 and 20% of the aggregate change in pay inequality in the sample period. Further, this operates entirely through within-industry channels; the between-industry response, if anything, works to reduce inequality over the nine-year change, though this effect is small. Consistent with this, in Appendix figure D.1 we show that the share of total payments accounted for by an industry does not respond to tariff liberalization.

[38] As our estimation reveals relative responses across industries, we cannot explain changes in pay inequality that are common across industries.

5 Pay Inequality Heterogeneity

The results above are unambiguous: tariff liberalization throughout the 1980s led to a reallocation within industries away from production workers towards non-production workers. However, the nature of the NBER-CES manufacturing data limits our ability to parse this result further. In particular, the data do not allow us to explore heterogeneity by worker or occupation types to more fully understand the labor market consequences of the liberalization.

To address this shortcoming, we turn to Census microdata from IPUMS (Ruggles et al., 2024). Specifically, we use the 5% 1980 and 1990 samples to calculate payments among employed individuals aged 25 to 65 for each of 72 Census industries. We define production workers as those belonging to the following occupation groups, as defined in Autor and Dorn (2013): production and craft workers; machine operators and assemblers; transport, construction, mechanical, mining, and farm workers. All other occupations are defined as non-production. These include clerical, administrative support and retail sales workers; managers, professional, and technical workers; and low-skill service workers.[39]

Table 6 about here.

We begin by replicating our baseline finding from column 4 of Table 3 in the IPUMS data. We report these results in column 1 of Table 6. The two samples yield similar findings: a one log point reduction in tariffs corresponds to a 3.2 log point increase in the log pay differential between production and non-production workers, compared to 2.7 in the NBER-CES data. In columns 2 and 3 of the table, we take advantage of the additional demographic information available in the IPUMS data to explore the effect separately for men and women. The results are striking: while the point estimate is negative for both groups, it is only statistically significant and is nearly three times as large for women.[40] That is, rising inequality in response to tariff liberalization seems to be driven overwhelmingly by women.[41]

[39] In appendix figure A.5 we show that the distribution of tariff changes and log pay inequality changes exhibit similar dispersion across the 4-digit 1987 SIC industries and census industries. Summary statistics for these variables can also be found in A.1.

[40] In figure A.6 we display the kernel density of changes in log pay inequality for men and women separately. Consistent with the summary statistics in table A.1 this distribution has a higher mean and variance for women than for men.

[41] In Appendix Table A.8 we present estimates in which we sequentially introduce controls as in Table 3. The differential effect among women is apparent in all specifications. Additionally, in Appendix Table A.9 we conduct "placebo" exercises and control for lagged outcomes as in Table 4. Finally, in Appendix Table A.10 we report similar results exploring changes in log payments to college graduates relative to non-college graduates. Here, too, we find an increase in inequality in response to tariff liberalization that is driven

These results suggest important differences in the nature of employment for men and women in this era. Figure 8 underscores this point. The figure displays the Herfindahl-Hirschman Index (HHI) of 1980 employment among men and women at the level of the Census industry, the Census occupation, and the occupation group, as defined above. For reference, the figure also displays the HHI implied by a uniform distribution at the various levels. As the figure shows, women's manufacturing employment is substantially more concentrated than men's employment in this era, suggesting that women may be particularly susceptible to shocks in particular industries and occupations.

Figure 8 about here.

This of course begs the question as to what kind of manufacturing jobs women are concentrated in. In this spirit, Table 7 lists the top 10 occupations by employment share among men and women, as well as the routineness index of the relevant occupation, as defined by Autor and Dorn (2013) and expressed in standard deviations relative to the mean. Six of the top seven occupations for women, accounting for more than 40% of women's employment, exhibit above mean routineness. Conversely, five of the top six occupations for men are below average in routineness.[42] Women's manufacturing employment in the era is highly skewed towards routine occupations such as textile sewing machine operators, equipment assemblers, and secretaries. Men, on the other hand, are much more likely to work as production supervisors, machine operators, and managers.

Table 7 about here.

Given the heterogeneity in the importance of routine occupations between men and women, in Table 8 we repeat the specifications from Table 6 separately for occupations above and below the median routineness index level. Here, too, we observe a striking pattern: among more routine occupations, there is a shift in pay towards non-production workers among both men and women. While the effect for women is larger than that for men, both are large and statistically significant. Conversely, neither group exhibits a statistically significant response among non-routine occupations. That is, the effects of tariff liberalization on inequality are driven exclusively by routine occupations, with an effect that falls on both men and women, albeit more heavily on women. This suggests a possible explanation for the stark heterogeneity in Table 6: women are disproportionately employed in routine oc-

entirely by women.

[42] At the occupation group level, approximately 80% of women's employment is accounted for by the three most routine occupation groups: machine operators, craft producers, and clerical workers. Conversely, nearly 50% of men's employment is accounted for by the three less routine occupation groups: transportation workers, managers, and low-skill service workers.

cupations in the era, and it is these occupations that experience the greatest rise in pay inequality in response to the Tokyo Round.

Table 8 about here.

In a similar spirit, we explore heterogeneity along another dimension: exposure to automation. Technological change is commonly cited as a potential driver of inequality in this era, and indeed its explanatory power is often directly contrasted with that of trade. The results above suggest that trade and technology may have jointly played a role in determining the shape of inequality in the era. In particular, it is possible that industries differentially exposed to technological change responded differently to trade liberalization. We explore this possibility more directly in Table 9, in which we repeat the specifications from Table 8 separately for industries above and below the median level of automation (Acemoglu and Restrepo, 2019, 2020).

Table 9 about here.

Again, we observe a clear pattern: among industries at or above the median level of automation between 1947 and 1978, the Tokyo Round liberalization dramatically increases pay differentials. Specifically, an interquartile increase in the liberalization raises pay inequality by 0.17 log points, with an effect that is statistically significant for both men and women. Conversely, among industries below the median level of automation we observe a statistically significant effect of the liberalization only among women, for whom the estimated effect is 40% lower than in high automation industries. This suggests that, rather than operating in isolation, tariff liberalization and technological change interact to drive inequality in this era.

These findings underscore a broader point: the effect of the Tokyo Round on inequality was not uniform, and was shaped jointly by both trade and technological change. Workers and industries exposed to both falling tariffs and shifting technology experienced larger increases in the relative pay of non-production workers, while those only exposed to tariff liberalization saw more muted effects. It was the combination of trade and technology, rather than either in isolation, that reshaped labor markets in the era.

6 Conclusion

Despite representing a tariff liberalization larger in magnitude than any of CUSFTA, NAFTA, or granting PNTR to China, the Tokyo Round of the GATT has received relatively little

empirical attention. In this paper, we exploit newly available data and a novel identification strategy to explore the trade and distributional consequences of the Tokyo Round. We show that the formula-based nature of Tokyo Round tariff negotiations, combined with historical 1930s tariffs, allows construction of a simple IV that circumvents standard concerns about endogenous tariff protection.

We use our measure to show that tariff cuts led to rising 1980s import growth and were associated with an increase in pay inequality between production and non-production workers that is overwhelmingly caused by changes in employment, rather than rising wage inequality. This effect is driven by routine occupations and industries most exposed to technological change. Strikingly, the increases in inequality fall disproportionately on women, who are over-represented in routine manufacturing jobs in this era.

The Tokyo Round represents an interesting laboratory for future work given the magnitude of the liberalization alongside the clean and transparent identification strategy provided by our Swiss IV. Indeed, this is an era of falling manufacturing employment, rising women's labor force participation, declining unionization, and enormous shifts between workers of differing education levels. Each of these trends is potentially related, either directly or indirectly, to the Tokyo Round tariff liberalization.

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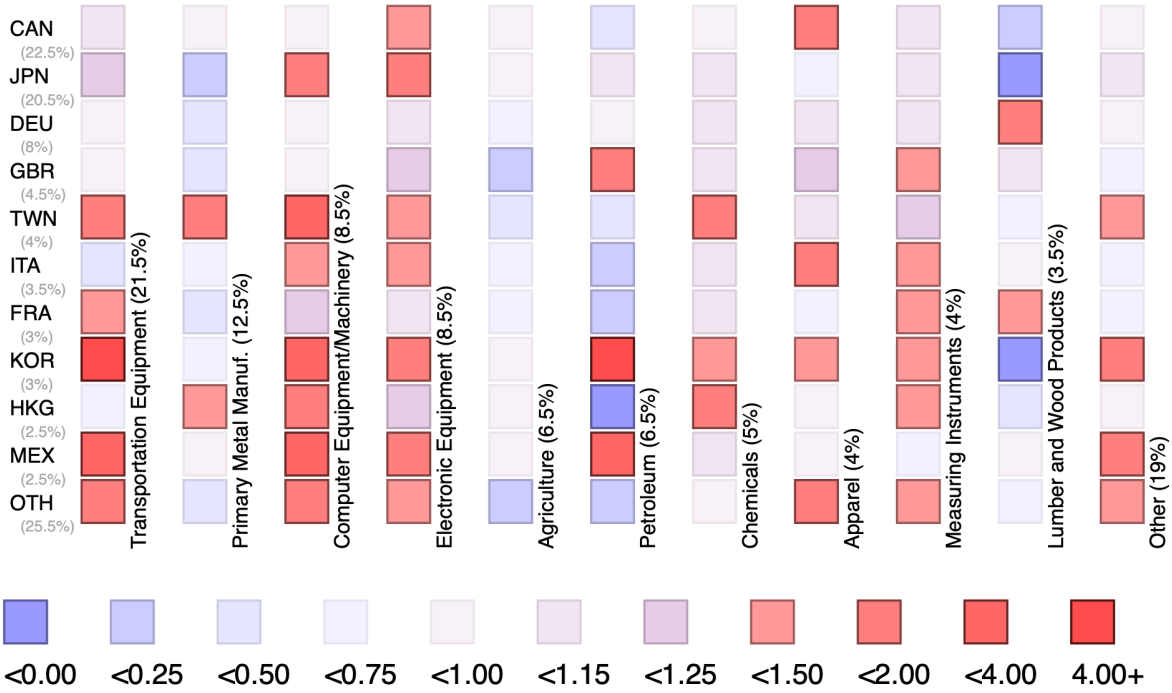
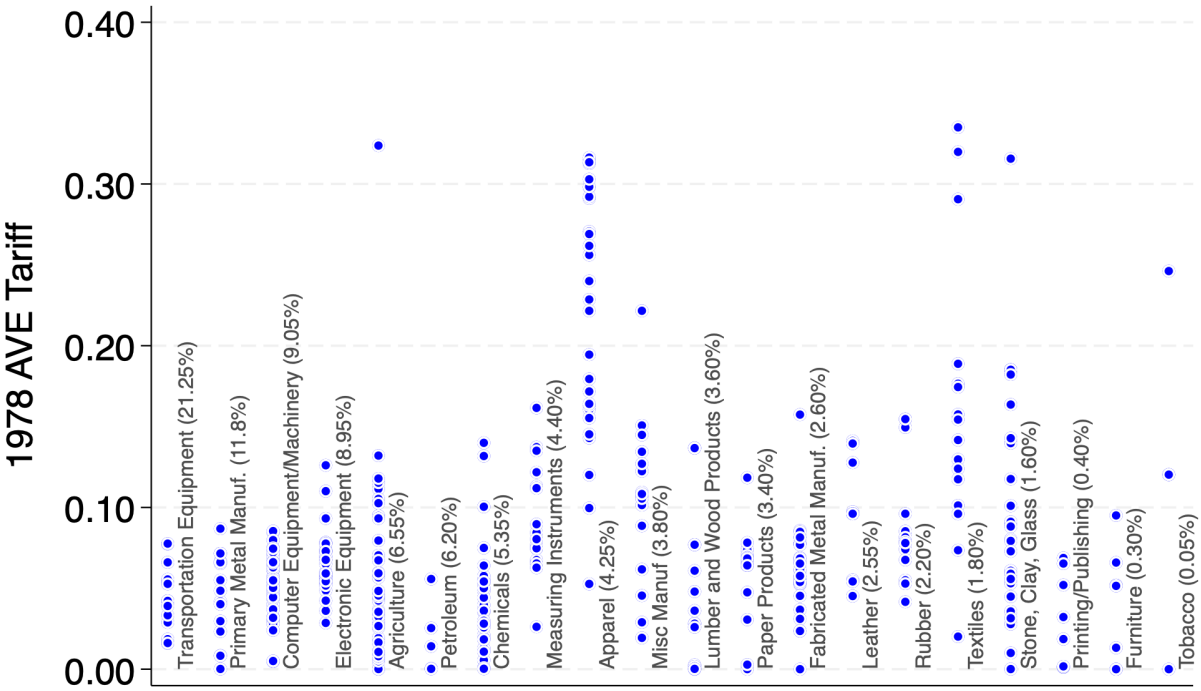
Tables and Figures

Figure 1: Import Growth by Country and Sector



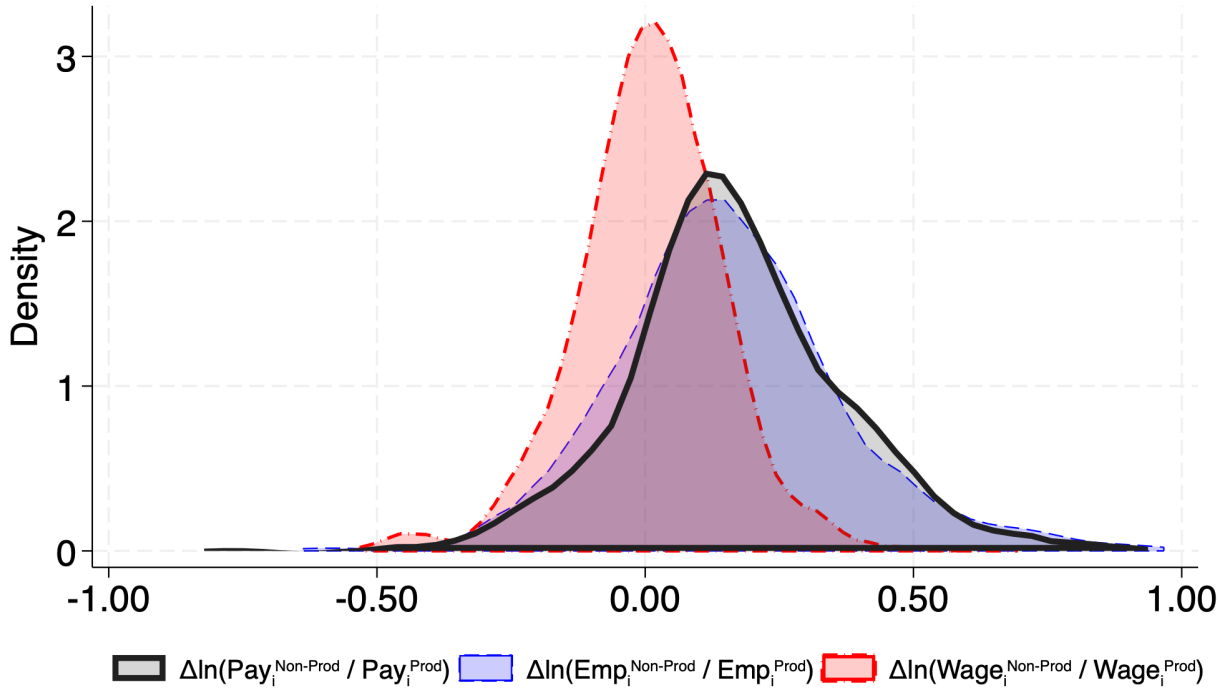
Notes: Each cell represents US log import growth between 1979 and 1988 from an exporting country in a 2-digit 1987 SIC sector. The numbers in parentheses on the vertical and horizontal axes are, respectively, shares of total US imports in 1978 accounted for by the exporter and 2-digit SIC sector. Import data taken from Schott (2008).

Figure 2: Pre-Tokyo 1978 AVE Tariffs by 2-digit Sector



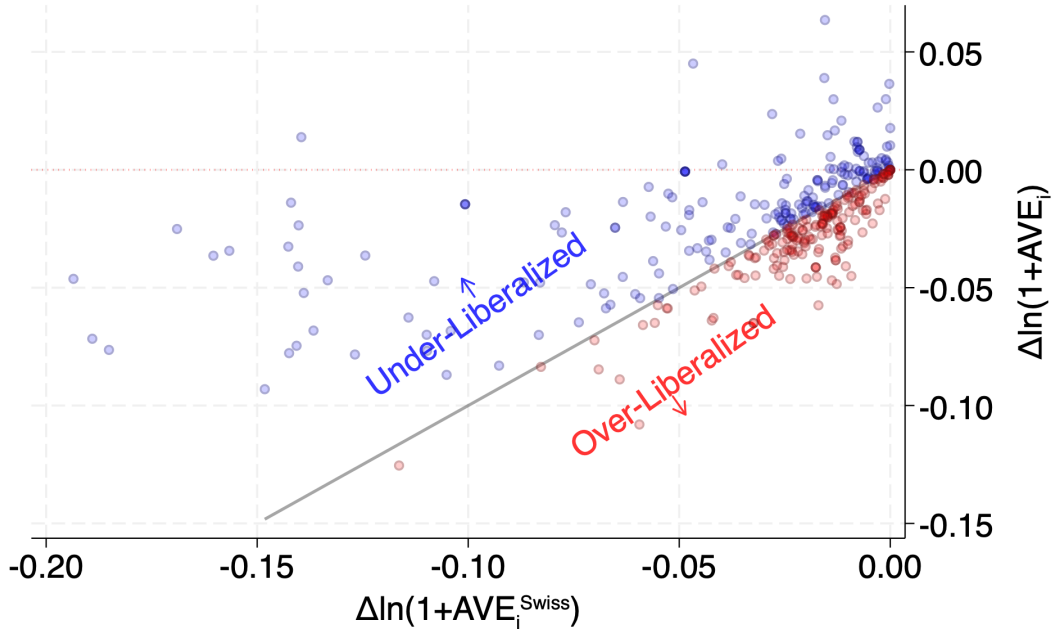
Notes: Each dot represents the 1978 AVE tariff of a 4-digit 1987 SIC industry. Dots clustered by two-digit SIC sectors and ordered by 1978 share of US imports, which are indicated in parentheses. Data from Greenland et al. (2025) and authors' calculations. Sample restricted to the 395 industries for which we are able to calculate imports, exports, and import penetration used in subsequent analysis. Sectors are ordered by 1978 import share (descending left to right). Import data are taken from Schott (2008).

Figure 3: Variation in Sources of Industry Inequality



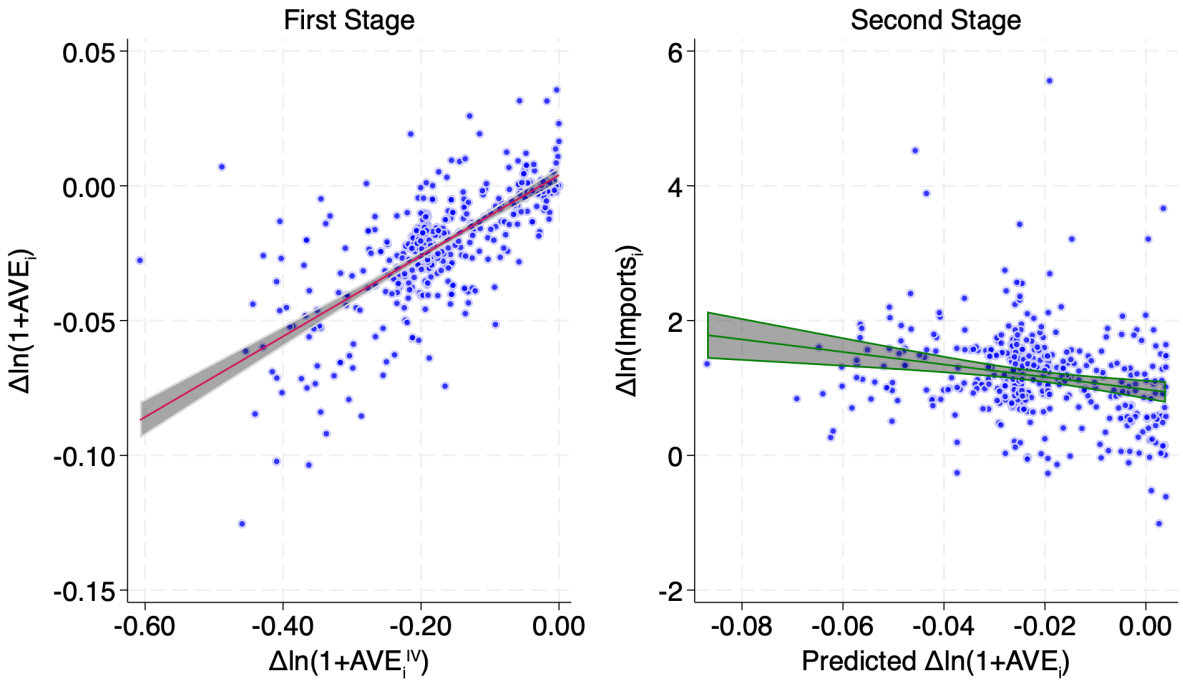
Notes: Figure displays a kernel density of the industry-level change in log pay to non-production (NP) workers relative to production (P) workers between 1979 and 1988 in thick black. Blue (with a thin dash) and red with a thick dash) densities decompose this log change into log changes in relative employment and relative wages respectively. Data from the NBER-CES manufacturing database, and authors' calculations.

Figure 4: AVE Tariff Cuts During the Tokyo Round



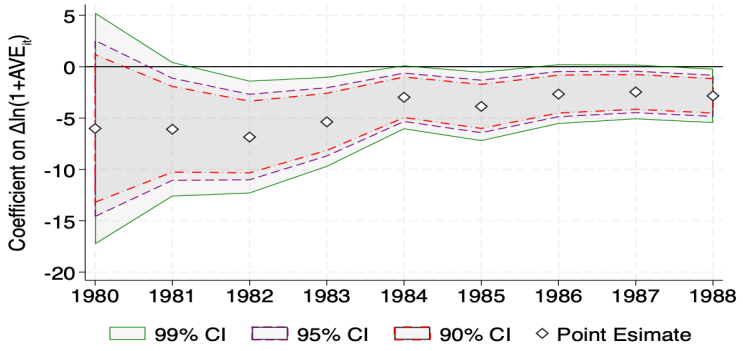
Notes: Figure displays a scatterplot of the 395 4-digit 1987 SIC industry-level tariff changes from 1979-1988 observed in practice (on the vertical axis) versus those implied by the Swiss formula (on the horizontal axis). The grey line indicates strict adherence to the Swiss formula. Points above this line (indicated in blue) received a smaller tariff cut than prescribed while those below the line (indicated in red) received a larger tariff cut than prescribed. Data taken from Greenland et al. (2025) and authors' calculations.

Figure 5: Import Growth Versus Instrumented Tariff Changes

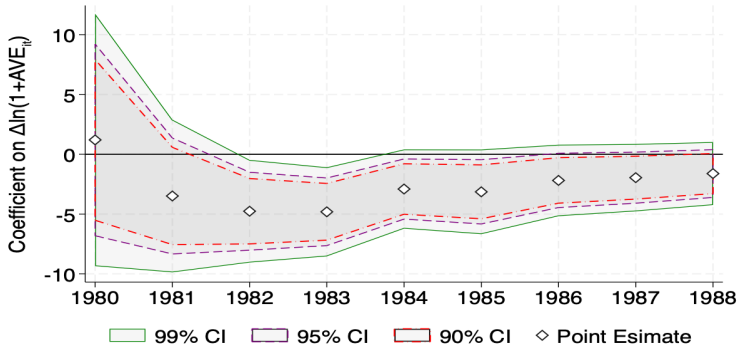


Notes: Scatterplot displaying two-stage least squares regression of import growth on changes in observed AVE tariffs $\Delta \ln(1+AVE_i)$ instrumented by our Swiss IV$\Delta \ln(1+AVE_i^{IV})$ as defined in Equations 3 and 4. Long differences taken from 1979-1988. Two points omitted from this figure for improved visibility. Results are not dependent on their omission. Full figure available upon request. Data from Schott (2008), Greenland et al. (2025) and authors' calculations.

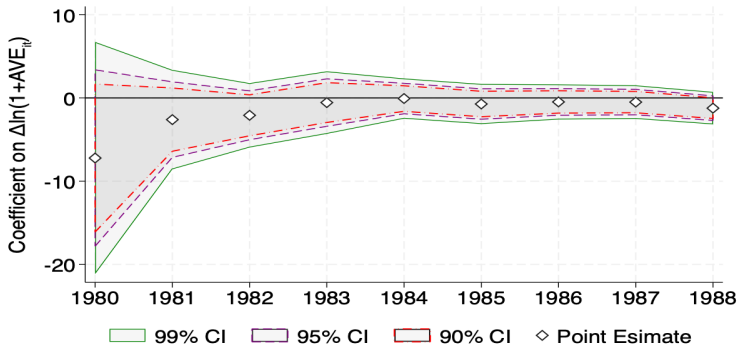
Figure 6: Decomposing Pay in to Employment and Wages



(a) $\Delta \ln\left(\frac{Pay^{Non-Prod}}{Pay^{Prod}}\right)$



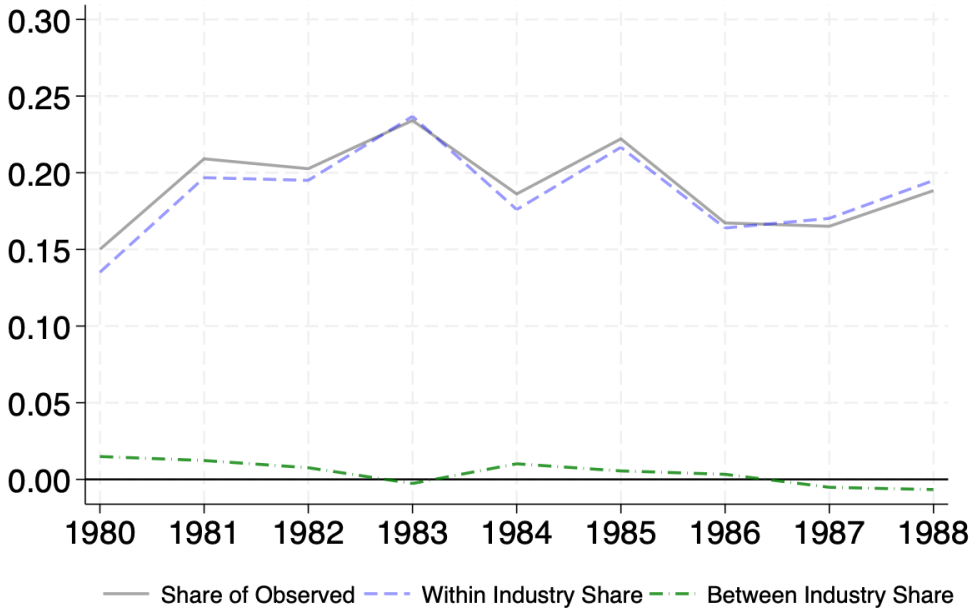
(b) $\Delta \ln\left(\frac{Emp^{Non-Prod}}{Emp^{Prod}}\right)$



(c) $\Delta \ln\left(\frac{Wage^{Non-Prod}}{Wage^{Prod}}\right)$

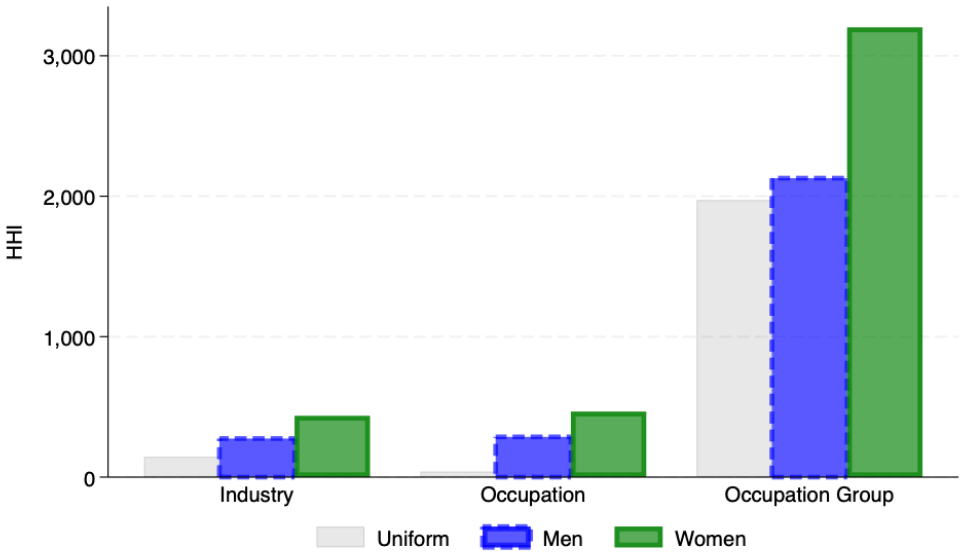
Notes: Figure presents estimates of equation 1 where the dependent variable is a change in log pay to non-production workers relative to production workers from from 1979-indicated year in Panel (a). Panels (b) and (c) decompose this effect into employment and wages. Specifications use full controls as in column 4 of Table 3.

Figure 7: Berman et al. (1994) Decomposition



Notes: Figure presents estimates of the effect of tariff liberalization on aggregate pay inequality as described in Berman et al. (1994). See the text for details on implementation and estimation in Appendix D.

Figure 8: Employment Concentration of Men and Women



Notes: Figure presents the HHI for men and women by employment at the level of the census industries, occupations, and occupation groups. Data from Ruggles et al. (2024).

Table 1: Correlates of Swiss IV

	$\Delta \ln(1 + AVE_i^{IV})$			
	(1)	(2)	(3)	(4)
$\Delta AVE_i^{Exports}$	0.0849***			0.0521***
	(0.0285)			(0.0167)
$\Delta \ln(p^*_{i,t-1})$	0.0882***			0.0366
	(0.0189)			(0.0230)
$I(MFA_i)$	-0.0948***			-0.0449***
	(0.0117)			(0.0134)
$STS_i * \frac{AVE_i}{1+AVE_i}$	-1.858***			-1.873***
	(0.315)			(0.233)
$\frac{Capital_i}{Labor_i}$		0.0176**		0.0243***
		(0.00785)		(0.00771)
$\frac{Emp_i^{Non-Prod}}{Emp_i}$		0.116**		0.0313
		(0.0504)		(0.0404)
$\frac{Emp_i^{Women}}{Emp_i}$		-0.244***		-0.0840**
		(0.0336)		(0.0389)
$\frac{Materials_i}{Shipments_i}$		0.172***		0.160***
		(0.0438)		(0.0327)
$\ln(Investment_i)$			0.0194*	0.00578
			(0.0104)	(0.0110)
$\Delta \ln(Investment_{i,t-1})$			-0.0179***	-0.00206
			(0.00538)	(0.00456)
$Routineness_i$			-0.0978***	-0.0193
			(0.0247)	(0.0196)
$Automation_i$			0.244***	0.0302
			(0.0693)	(0.0585)
Constant	-0.119***	-0.256***	-0.0471	-0.245***
	(0.00975)	(0.0338)	(0.0349)	(0.0510)
Observations	395	395	395	395
R^2	0.340	0.357	0.142	0.482

Notes: Table presents correlation between industry characteristics and our instrument $\Delta \ln(1+AVE_i^{IV})$ based on the formula outlined in equation 3. All data measured at or concorded to the 4-digit 1987 SIC classification. Robust standard errors reported in parentheses. *, **, and *** indicate p-values less than 10%, 5% and 1%, respectively. Changes are contemporaneous to the outcome unless the variable is subscripted $t-1$ indicating a change from 1972-1979.

Table 2: Tariff Liberalization and Trade

	(1) $\Delta \ln(Imp_i)$	(2) $\Delta \ln(Imp_i)$	(3) $\Delta \ln(Imp_i)$	(4) $\Delta \ln(Imp_i)$	(5) $\Delta ImpPen_i$	(6) $\Delta \ln(Exp_i)$	(7) $\Delta \frac{Exp_i}{Ship_i}$
$\Delta \ln(1 + AVE_i)$	-1.683	-8.968***	-9.661***	-8.788***	-1.562***	-3.548	-0.333
	(3.480)	(2.573)	(2.902)	(3.203)	(0.335)	(2.173)	(0.277)
$\Delta \ln(p^*_{i,t-1})$			-0.255**	-0.249**	0.00335	-0.154	-0.0214
			(0.126)	(0.127)	(0.0165)	(0.124)	(0.0254)
$STS_i * \frac{AVE_i}{1+AVE_i}$			-2.023	-1.271	-0.846***	1.671	-0.0764
			(1.779)	(1.961)	(0.198)	(1.450)	(0.0959)
$I(MFA_i)$				-0.0447	0.0173	-0.0904	-0.0122*
				(0.0852)	(0.0146)	(0.0737)	(0.00724)
$\Delta AVE_i^{Exports}$				-0.264	0.0253	0.0753	0.00299
				(0.251)	(0.0168)	(0.0909)	(0.0101)
Constant	1.122***	0.963***	0.979***	0.927***	0.0425***	0.446***	0.00977
	(0.105)	(0.0830)	(0.0849)	(0.107)	(0.00767)	(0.0505)	(0.00648)
IQE	-0.071	-0.263	-0.283	-0.258	-0.046	-0.104	-0.010
Estimator	OLS	IV	IV	IV	IV	IV	IV
Obs.	392	392	392	392	392	392	392
1st Stage Coeff.		0.149	0.150	0.143	0.143	0.143	0.143
Under ID p-stat	.	0.000	0.000	0.000	0.000	0.000	0.000
KP F-Stat	.	177.841	153.342	115.979	115.979	115.979	115.979

Notes: Table presents estimates of equation 1 where the dependent variable is a change in log imports in columns 1-4, change in import penetration, which is imports divided by shipments minus exports plus imports, in column 5, change in log exports in column 5, and change in exports scaled by shipments in column 7. AVE tariffs have been constructed at the SIC87 level with data from Greenland et al. (2025) and are instrumented in columns 2-7 with the Swiss formula implied tariff change as in equations 3 and 4. All data measured at or concorded to the 4-digit 1987 SIC classification. First stage coefficients, under-identified p-value and Kleibergen-Paap Rk F-statistic are reported in the footer of the table. IQE indicates the effect on the dependent variable an interquartile increase in $\Delta \ln(1 + AVE_i)$. Robust standard errors reported in parentheses. *, **, and *** indicate p-values less than 10%, 5% and 1%, respectively.

Table 3: Tariff Liberalization and Pay Inequality

	$\Delta \ln(\frac{Pay_i^{Non-Prod}}{Pay_i^{Prod}})$			
	(1)	(2)	(3)	(4)
$\Delta \ln(1 + AVE_i)$	-1.804***	-2.994***	-2.839***	-2.838***
	(0.691)	(0.824)	(1.012)	(1.000)
$\Delta \ln(p^*_{i,t-1})$		-0.0375	-0.0377	-0.0385
		(0.0460)	(0.0469)	(0.0467)
$STS_i * \frac{AVE_i}{1+AVE_i}$		-0.605	-0.423	-0.143
		(0.572)	(0.624)	(0.630)
$I(MFA_i)$		-0.0755***	-0.0847***	-0.0795**
		(0.0285)	(0.0300)	(0.0309)
$\Delta AVE_i{}^{Exports}$		0.0331	0.0334	0.0480
		(0.0367)	(0.0358)	(0.0344)
$\frac{Capital_i}{Labor_i}$			-0.0226	-0.00372
			(0.0198)	(0.0218)
$\frac{Emp_i^{Non-Prod}}{Emp_i}$			0.0222	0.0187
			(0.113)	(0.114)
$\frac{Emp_i^{Women}}{Emp_i}$			-0.0301	0.0316
			(0.0999)	(0.110)
$\frac{Materials_i}{Shipments_i}$			0.0703	0.0798
			(0.115)	(0.114)
$\ln(Investment_i)$				-0.0122
				(0.0223)
$\Delta \ln(Investment_{i,t-1})$				0.0131
				(0.0131)
$Routineness_i$				0.0211
				(0.0463)
$Automation_i$				0.318**
				(0.159)
Constant	0.127***	0.133***	0.194*	0.00313
	(0.0191)	(0.0210)	(0.102)	(0.139)
IQE	-0.053	-0.088	-0.084	-0.084
Obs.	395	395	395	395
1st Stage Coeff.	0.150	0.144	0.135	0.136
Under ID p-stat	0.000	0.000	0.000	0.000
KP F-Stat	180.332	119.386	93.062	89.761
Estimator	IV	IV	IV	IV
Controls	None	+Trade	+Prod.	+Tech.

Notes: Table presents estimates of equation 1 where the dependent variable is the change in log pay to non-production workers relative to production workers from 1979-1988. AVE tariffs have been constructed at the SIC87 level with data from Greenland et al. (2025) and are instrumented with the Swiss formula implied tariff change as in equations 3 and 4. All data measured at or concorded to the 4-digit 1987 SIC classification. New controls are added sequentially and indicated in the footer. First stage coefficients, under-identified p-value and Kleibergen-Paap Rk F-statistic are reported in the footer of the table. IQE indicates the effect on the dependent variable of an interquartile increase in $\Delta \ln(1 + AVE_i)$. Robust standard errors reported in parentheses. *, **, and *** indicate p-values less than 10%, 5% and 1%, respectively.

Table 4: Tariff Liberalization and Pay Inequality Pre-Trends

	(1)	(2)	(3)	(4)
	Panel A: $\Delta \ln(\frac{Pay_i^{Non-Prod}}{Pay_i^{Prod}})_{t-1}$			
$\Delta \ln(1 + AVE_i)$	0.187	1.238	1.921**	1.827*
	(0.522)	(0.620)	(0.741)	(0.736)
IQE	0.004	0.028	0.044	0.042
Obs.	394	394	394	394
1st Stage Coeff.	0.150	0.144	0.135	0.136
Under ID p-stat	0.000	0.000	0.000	0.000
KP F-Stat	180.335	119.304	93.063	89.762
	Panel B: $\Delta \ln(\frac{Pay_i^{Non-Prod}}{Pay_i^{Prod}})_{t}$			
$\Delta \ln(1 + AVE_i)$	-1.764***	-2.756***	-2.446**	-2.458**
	(0.682)	(0.811)	(1.010)	(0.992)
$\Delta \ln(\frac{Pay_i^{Non-Prod}}{Pay_i^{Prod}})_{t-1}$	-0.212**	-0.197**	-0.204**	-0.207**
	(0.124)	(0.123)	(0.124)	(0.123)
IQE	-0.052	-0.081	-0.072	-0.072
Obs.	394	394	394	394
1st Stage Coeff.	0.150	0.144	0.135	0.136
Under ID p-stat	0.000	0.000	0.000	0.000
KP F-Stat	177.851	115.526	88.769	86.250
Estimator	IV	IV	IV	IV
Controls	None	+Trade	+Prod.	+Tech.

Notes: Table presents estimates of equation 1 where the dependent variable is the change in log pay to non-production workers relative to production workers. Panel A estimates this relationship between 1972-1979 as a placebo exercise. Panel B re-estimates table 3 but controls for the lagged change from 1972-1979. AVE tariffs have been constructed at the SIC87 level with data from Greenland et al. (2025) and are instrumented with the Swiss formula implied tariff change as in equations 3 and 4. All data measured at or concorded to the 4-digit 1987 SIC classification. New controls are added sequentially and indicated in the footer. First stage coefficients, under-identified p-value and Kleibergen-Paap Rk F-statistic are reported in the footer of the table. IQE indicates the effect on the dependent variable of an interquartile increase in $\Delta \ln(1 + AVE_i)$. Robust standard errors reported in parentheses. *, **, and *** indicate p-values less than 10%, 5% and 1%, respectively.

Table 5: Estimating the Effects of Upstream Tariffs on Inequality

	$\Delta \ln(\frac{Pay_i^{Non-Prod}}{Pay_i^{Prod}})$			
	(1)	(2)	(3)	(4)
$\Delta \ln(1 + AVE_i)$	-1.628**	-2.426***	-2.574**	-2.503**
	(0.765)	(0.872)	(1.006)	(1.000)
$\Delta \ln\left(1 + AVE_i^{Up}\right)$	-1.138	-5.222*	-7.432**	-7.610*
	(2.893)	(2.957)	(3.784)	(3.952)
IQE	-0.048	-0.071	-0.076	-0.074
IQE^{up}	-0.006	-0.030	-0.042	-0.043
Obs.	394	394	394	394
Under ID p-stat	0.000	0.000	0.000	0.000
KP F-Stat	76.007	55.748	45.244	42.843
Estimator	IV	IV	IV	IV
Controls	None	+Trade	+Prod.	+Tech.

Notes: Table presents estimates of equation 1 where the dependent variable is the change in log pay to non-production workers relative to production workers from 1979-1988. AVE tariffs have been constructed at the SIC87 level with data from Greenland et al. (2025) and are instrumented with the Swiss formula implied tariff change as in equations 3 and 4. All data measured at or concorded to the 4-digit 1987 SIC classification. New controls are added sequentially and indicated in the footer. First stage coefficients, under-identified p-value and Kleibergen-Paap Rk F-statistic are reported in the footer of the table. IQE indicates the effect on the dependent variable of an interquartile increase in $\Delta \ln(1+AVE_i)$. Robust standard errors reported in parentheses. *, **, and *** indicate p-values less than 10%, 5% and 1%, respectively.

Table 6: Heterogeneous Effects of Tariff Liberalization for Men and Women

	$\Delta \ln(\frac{Pay_i^{Non-Prod}}{Pay_i^{Prod}})$		
	All	Men	Women
$\Delta \ln(1 + AVE_i)$	-3.270*	-2.475	-6.783***
	(1.693)	(1.990)	(2.093)
$\Delta \ln(p^*_{i,t-1})$	0.210	0.171	0.193
	(0.224)	(0.245)	(0.260)
$STS_i * \frac{AVE_i}{1+AVE_i}$	-0.842	-0.0845	-0.579
	(1.243)	(1.634)	(1.802)
$I(MFA_i)$	-0.0583	-0.0675	-0.0401
	(0.0439)	(0.0508)	(0.0549)
$\Delta AVE_i^{Exports}$	0.163	0.0565	0.304*
	(0.134)	(0.164)	(0.161)
$\frac{Capital_i}{Labor_i}$	0.00754	-0.00753	0.0656
	(0.0407)	(0.0483)	(0.0678)
$\frac{Emp_i^{Non-Prod}}{Emp._i}$	0.153	0.0874	0.512*
	(0.168)	(0.178)	(0.284)
$\frac{Emp_i^{Women}}{Emp_i}$	0.358	0.0832	0.851***
	(0.227)	(0.240)	(0.324)
$\frac{Materials_i}{Shipments_i}$	-0.0959	-0.105	-0.0129
	(0.113)	(0.114)	(0.240)
$\ln(Investment_i)$	0.00758	0.00367	-0.0732
	(0.0299)	(0.0325)	(0.0518)
$\Delta \ln(Investment)_{i,t-1}$	-0.0104	-0.0251	0.0573
	(0.0277)	(0.0306)	(0.0369)
$Routineness_i$	-0.0162	0.0223	0.0176
	(0.0577)	(0.0645)	(0.0742)
$Automation_i$	0.166	0.0721	0.355
	(0.226)	(0.254)	(0.259)
Constant	0.0937	0.184	-0.534
	(0.230)	(0.272)	(0.340)
IQE	-0.090	-0.068	-0.187
Obs.	72	72	72
1st Stage Coeff.	0.130	0.130	0.130
Under ID p-stat	0.000	0.000	0.000
KP F-Stat	32.136	32.136	32.136
Estimator	IV	IV	IV
Controls	Full	Full	Full

Notes: Table presents estimates of equation 1 where the dependent variable is the change in log pay to non-production workers relative to production workers from 1980-1990 among all workers (column 1), men (column 2), and women (column 3). AVE tariffs from Greenland et al. (2025) and are instrumented with the Swiss formula implied tariff change as in equations 3 and 4. Outcome data taken from Ruggles et al. (2024). Data measured at or concorded to the census industry classification. First stage coefficients, under-identified p-value and Kleibergen-Paap Rk F-statistic are reported in the footer of the table. IQE indicates the effect on the dependent variable of an interquartile increase in $\Delta \ln(1 + AVE_i)$. Robust standard errors reported in parentheses. *, **, and *** indicate p-values less than 10%, 5% and 1%, respectively.

Table 7: Men's and Women's Employment Share

Occupation	Routineness	Emp Share
	Panel A: Women	
Sewing Machine Operators	0.09	0.11
Electrical Equipment Assemblers	0.92	0.10
Machine Operators	-0.12	0.09
Secretaries	2.54	0.08
Graders and Sorters	0.41	0.06
Bookkeepers	2.02	0.03
Packers and Packagers	0.50	0.03
Production Supervisors	-0.90	0.03
Office Clerks	1.54	0.02
Managers	-1.02	0.02
	Panel B: Men	
Production Supervisors	-0.90	0.09
Machine Operators	-0.12	0.08
Managers	-1.02	0.07
Truck and Tractor Drivers	-1.03	0.04
Electrical Equipment Assemblers	0.92	0.04
Freight and Materials Handlers	-0.24	0.03
Welders and Metal Cutters	0.26	0.03
Machinists	0.37	0.03
Graders and Sorters	0.41	0.03
Industrial Machinery Repairers	-0.34	0.02

Notes: Table presents the top 10 occupations by employment share and sex by routineness index (Autor and Dorn, 2013).

Table 8: Tariff Liberalization and Inequality by Routine Status

	$\Delta \ln(\frac{Pay_i^{Non-Prod}}{Pay_i^{Prod}})$		
	All	Men	Women
	Panel (a): Routine Occupations		
$\Delta \ln(1 + AVE_i)$	-4.864**	-4.164*	-8.235***
	(1.957)	(2.211)	(2.275)
IQE	-0.134	-0.115	-0.228
	Panel (b): Non-Routine Occupations		
$\Delta \ln(1 + AVE_i)$	-1.374	-1.245	-1.475
	(2.018)	(2.267)	(3.250)
IQE	-0.038	-0.034	-0.041
Obs.	72	72	72
1st Stage Coeff.	0.130	0.130	0.130
Under ID p-stat	0.000	0.000	0.000
KP F-Stat	32.136	32.136	32.136
Specification	IV	IV	IV
Controls	Full	Full	Full

Notes: Table presents estimates of equation 1 where the dependent variable is the change in log pay to non-production workers relative to production workers from 1980-1990 among all workers (column 1), men (column 2), and women (column 3). Panel (a) presents estimates among occupations with above median routineness, while Panel (b) presents estimates for below median routineness occupations. AVE tariffs from Greenland et al. (2025) and are instrumented with the Swiss formula implied tariff change as in equations 3 and 4. Outcome data taken from Ruggles et al. (2024). Data measured at or concorded to the census industry classification. First stage coefficients, under-identified p-value and Kleibergen-Paap Rk F-statistic are reported in the footer of the table. IQE indicates the effect on the dependent variable of an interquartile increase in $\Delta \ln(1 + AVE_i)$. Robust standard errors reported in parentheses. *, **, and *** indicate p-values less than 10%, 5% and 1%, respectively.

Table 9: Tariff Liberalization and Inequality by Automation

	$\Delta \ln(\frac{Pay_i^{Non-Prod}}{Pay_i^{Prod}})$		
	All	Men	Women
	Panel (a): High Automation		
$\Delta \ln(1 + AVE_i)$	-6.129***	-5.697**	-6.674**
	(2.157)	(2.736)	(2.865)
IQE	-0.169	-0.157	-0.184
Obs.	38	38	38
1st Stage Coeff.	0.164	0.164	0.164
Under ID p-stat	0.004	0.004	0.004
KP F-Stat	12.848	12.848	12.848
	Panel (b): Low Automation		
$\Delta \ln(1 + AVE_i)$	-1.272	-0.710	-4.580**
	(2.081)	(2.434)	(2.212)
IQE	-0.035	-0.020	-0.127
Obs.	34	34	34
1st Stage Coeff.	0.128	0.128	0.128
Under ID p-stat	0.005	0.005	0.005
KP F-Stat	6.399	6.399	6.399
Specification	IV	IV	IV
Controls	Full	Full	Full

Notes: Table presents estimates of equation 1 where the dependent variable is the change in log pay to non-production workers relative to production workers from 1980-1990 among all workers (column 1), men (column 2), and women (column 3). Panel (a) presents estimates among industries with above median automation, while Panel (b) presents estimates for below median automation industries occupations. AVE tariffs from Greenland et al. (2025) and are instrumented with the Swiss formula implied tariff change as in equations 3 and 4. Outcome data taken from Ruggles et al. (2024). Data measured at or concorded to the census industry classification. First stage coefficients, under-identified p-value and Kleibergen-Paap Rk F-statistic are reported in the footer of the table. IQE indicates the effect on the dependent variable of an interquartile increase in $\Delta \ln(1 + AVE_i)$. Robust standard errors reported in parentheses. *, **, and *** indicate p-values less than 10%, 5% and 1%, respectively.

Appendix

A Appendix Tables and Figures

Table A.1: Summary Statistics

	Obs.	μ	σ	IQR
	SIC Analysis			
$\Delta \ln\left(1 + AVE_i^{IV}\right)$	395	-0.174	0.109	0.135
$\Delta \ln\left(1 + AVE_i\right)$	395	-0.022	0.023	0.029
$\Delta \frac{Exports_{i,t}}{Shipments_{i,t}}$	395	0.012	0.06	0.027
ΔImport Penetration$_i$	395	0.066	0.086	0.084
$\Delta \ln\left(Exports_i\right)$	393	0.492	0.507	0.629
$\Delta \ln\left(Imports_i\right)$	394	1.166	0.846	0.693
$\Delta \ln(\frac{Pay_i^{Non-Prod}}{Pay_i^{Prod}})$	395	0.166	0.213	0.241
$\Delta \ln(\frac{Emp_i^{Non-Prod}}{Emp_i^{Prod}})$	395	0.156	0.210	0.250
$\Delta \ln(\frac{Wage_i^{Non-Prod}}{Wage_i^{Prod}})$	395	0.010	0.139	0.161
	Census Industry Analysis			
$\Delta \ln\left(1 + AVE_i^{IV}\right)$	72	-0.153	0.094	0.140
$\Delta \ln\left(1 + AVE_i\right)$	72	-0.020	0.02	0.028
$\Delta \ln(\frac{Pay_i^{Non-Prod,A}}{Pay_i^{Prod,A}})$	72	0.231	0.147	0.174
$\Delta \ln(\frac{Pay_i^{Non-Prod,M}}{Pay_i^{Prod,M}})$	72	0.193	0.157	0.181
$\Delta \ln(\frac{Pay_i^{Non-Prod,W}}{Pay_i^{Prod,W}})$	72	0.288	0.254	0.332

Notes: Summary statistics of key variables. Panel A reports descriptive statistics aggregated to the 4-digit 1987 SIC industry. Panel B reports descriptive statistics for data measured at or concorded to the census industries among traded sectors as obtained from IPUMS Ruggles et al. (2024). Import data are taken from Schott (2008), AVE reflect authors' calculations based on data in Greenland et al. (2025), remaining data are taken from the NBER Manufacturing Database and Ruggles et al. (2024). $\Delta \ln(1 + AVE_i)$ is the change in the log of one plus the observed ad valorem equivalent tariff. $\Delta \ln(1 + AVE_i^{IV})$ is the change in the log of one plus the AVE tariff implied by the Swiss formula defined and the column 2 tariff rate as in equation (2).

Table A.2: Placebo Tariff Liberalization and Trade – 1972-1979

	(1) $\Delta\ln(Imp_i)$	(2) $\Delta\ln(Imp_i)$	(3) $\Delta\ln(Imp_i)$	(4) $\Delta\ln(Imp_i)$	(5) $\Delta ImpPen_i$	(6) $\Delta\ln(Exp_i)$	(7) $\Delta\frac{Exp_i}{Ship_i}$
$\Delta\ln(1+AVE_i)$	2.307	4.934	2.785	0.525	-0.289	-2.772	0.197
	(2.040)	(3.462)	(3.339)	(3.309)	(0.383)	(3.155)	(0.414)
$\Delta\ln(p^*_{i,t-1})$			-1.038***	-0.989***	-0.00136	0.357	0.000937
			(0.350)	(0.356)	(0.0126)	(0.230)	(0.0202)
$STS_i * \frac{AVE_i}{1+AVE_i}$			-6.656**	-7.189***	-0.350**	-2.585	-0.226
			(2.727)	(2.756)	(0.173)	(2.236)	(0.165)
$I(MFA_i)$				-0.0936	0.0274**	0.169*	0.000216
				(0.132)	(0.0127)	(0.100)	(0.00674)
$\Delta AVE_i{}^{Exports}$				0.236	0.0239*	0.207	-0.0000546
				(0.187)	(0.0136)	(0.170)	(0.00911)
Constant	1.541***	1.599***	1.663***	1.700***	0.0323***	1.690***	0.0446***
	(0.0637)	(0.0831)	(0.0871)	(0.0968)	(0.0100)	(0.0896)	(0.0124)
IQE	0.098	0.145	0.082	0.015	-0.008	-0.081	0.006
Estimator	OLS	IV	IV	IV	IV	IV	IV
Obs.	389	389	389	389	392	390	392
1st Stage Coeff.		0.149	0.150	0.143	0.143	0.143	0.143
Under ID p-stat	.	0.000	0.000	0.000	0.000	0.000	0.000
KP F-Stat	.	176.813	151.711	113.723	115.979	111.301	115.979

Notes: Table presents placebo estimates of equation 1 where the dependent variable is the change taken from 1972-1979 in log imports in columns 1-4, change in import penetration, which is imports divided by shipments minus exports plus imports, in column 5, change in log exports in column 5, and change in exports scaled by shipments in column 7. To facilitate comparison with primary estimates these changes have been made into 9-year equivalent changes. AVE tariffs have been constructed at the SIC87 level with data from Greenland et al. (2025) and are instrumented in columns 2-7 with the Swiss formula implied tariff change as in equations 3 and 4. All data measured at or concorded to the 4-digit 1987 SIC classification. First stage coefficients, under-identified p-value and Kleibergen-Paap Rk F-statistic are reported in the footer of the table. IQE indicates the effect on the dependent variable an interquartile increase in $\Delta\ln(1+AVE_i)$. Robust standard errors reported in parentheses. *, **, and *** indicate p-values less than 10%, 5% and 1%, respectively.

Table A.3: Tariff Liberalization and Trade – Alternate IV

	(1) $\Delta\ln(Imp_i)$	(2) $\Delta\ln(Imp_i)$	(3) $\Delta\ln(Imp_i)$	(4) $\Delta\ln(Imp_i)$	(5) $\Delta ImpPen_i$	(6) $\Delta\ln(Exp_i)$	(7) $\Delta\frac{Exp_i}{Ship_i}$
$\Delta\ln(1+AVE_i)$	-1.683	-8.714***	-9.386***	-8.486***	-1.567***	-3.586	-0.355
	(3.480)	(2.527)	(2.859)	(3.197)	(0.332)	(2.182)	(0.275)
$\Delta\ln(p^*_{i,t-1})$			-0.259**	-0.254**	0.00344	-0.153	-0.0210
			(0.126)	(0.127)	(0.0164)	(0.124)	(0.0254)
$STS_i * \frac{AVE_i}{1+AVE_i}$			-1.937	-1.192	-0.847***	1.661	-0.0824
			(1.776)	(1.961)	(0.196)	(1.440)	(0.0950)
$I(MFA_i)$				-0.0402	0.0173	-0.0910	-0.0125*
				(0.0856)	(0.0147)	(0.0737)	(0.00730)
$\Delta AVE_i^{Exports}$				-0.269	0.0254	0.0760	0.00341
				(0.253)	(0.0168)	(0.0912)	(0.0101)
Constant	1.122***	0.969***	0.984***	0.930***	0.0425***	0.445***	0.00949
	(0.105)	(0.0816)	(0.0836)	(0.106)	(0.00768)	(0.0507)	(0.00646)
IQE	-0.049	-0.256	-0.275	-0.249	-0.046	-0.105	-0.010
Estimator	OLS	IV	IV	IV	IV	IV	IV
Obs.	392	392	392	392	392	392	392
1st Stage Coeff.		-0.093	-0.094	-0.089	-0.089	-0.089	-0.089
Under ID p-stat	.	0.000	0.000	0.000	0.000	0.000	0.000
KP F-Stat	.	169.781	143.742	110.818	110.818	110.818	110.818

Notes: Table presents estimates of equation 1 where the dependent variable is the change taken from 1979-1988 in log imports in columns 1-4, change in import penetration, which is imports divided by shipments minus exports plus imports, in column 5, change in log exports in column 5, and change in exports scaled by shipments in column 7. AVE tariffs have been constructed at the SIC87 level with data from Greenland et al. (2025) and are instrumented in columns 2-7 with the column 2 implied initial tariff level. All data measured at or concorded to the 4-digit 1987 SIC classification. First stage coefficients, under-identified p-value and Kleibergen-Paap Rk F-statistic are reported in the footer of the table. IQE indicates the effect on the dependent variable an interquartile increase in $\Delta\ln(1+AVE_i)$. Robust standard errors reported in parentheses. *, **, and *** indicate p-values less than 10%, 5% and 1%, respectively.

Table A.4: Tariff Liberalization and Trade Outcomes – Controlling for Lagged Changes

	(1) $\Delta \ln(Imp_i)$	(2) $\Delta \ln(Imp_i)$	(3) $\Delta \ln(Imp_i)$	(4) $\Delta \ln(Imp_i)$	(5) $\Delta ImpPen_i$	(6) $\Delta \ln(Exp_i)$	(7) $\Delta \frac{Exp_i}{Ship_i}$
$\Delta \ln(1 + AVE_i)$	-1.683	-9.529***	-10.80***	-10.11***	-1.527***	-3.340	-0.294
	(3.480)	(2.202)	(2.361)	(2.840)	(0.368)	(2.190)	(0.258)
$\Delta \ln(Imp_{i,t-1})$		-0.118*	-0.148**	-0.144**			
		(0.0692)	(0.0718)	(0.0706)			
$\Delta \text{ImpPen}_{i,t-1}$					0.123		
					(0.237)		
$\Delta \ln(Exp_{i,t-1})$						-0.0782*	
						(0.0440)	
$\Delta \frac{Exp_i}{Ship_i}_{t-1}$							-0.195***
							(0.0714)
IQE	-0.071	-0.280	-0.317	-0.297	-0.045	-0.098	-0.009
Estimator	OLS	IV	IV	IV	IV	IV	IV
Obs.	392	389	389	389	392	390	392
1st Stage Coeff.		0.148	0.148	0.141	0.143	0.142	0.143
Under ID p-stat	.	0.000	0.000	0.000	0.000	0.000	0.000
KP F-Stat	.	175.892	150.757	113.460	116.828	110.841	115.170

Notes: Table presents estimates of equation 1 where the dependent variable is a change taken from 1979-1988 in log imports in columns 1-4, change in import penetration, which is imports divided by shipments minus exports plus imports, in column 5, change in log exports in column 5, and change in exports scaled by shipments in column 7. As an additional control we include the lagged (1972-1979) change in the dependent variable. To facilitate comparison with primary estimates these lagged changes have been made into 9-year equivalent changes. AVE tariffs have been constructed at the SIC87 level with data from Greenland et al. (2025) and are instrumented in columns 2-7 with the Swiss formula implied tariff change as in equations 3 and 4. All data measured at or concorded to the 4-digit 1987 SIC classification. First stage coefficients, under-identified p-value and Kleibergen-Paap Rk F-statistic are reported in the footer of each table. IQE indicates the effect on the dependent variable an interquartile increase in $\Delta \ln(1 + AVE_i)$. Robust standard errors reported in parentheses. *, **, and *** indicate p-values less than 10%, 5% and 1%, respectively.

Table A.5: Estimating the Effects of Upstream Tariffs on Trade Flows

	(1) $\Delta\ln(Imp_i)$	(2) $\Delta\ln(Imp_i)$	(3) $\Delta\ln(Imp_i)$	(4) $\Delta\ln(Imp_i)$	(5) $\Delta ImpPen_i$	(6) $\Delta\ln(Exp_i)$	(7) $\Delta\frac{Exp_i}{Ship_i}$
$\Delta\ln(1+AVE_i)$	0.200	-7.458***	-7.905***	-7.000**	-1.069***	-3.183	-0.298
	(3.701)	(2.674)	(3.000)	(3.294)	(0.345)	(2.374)	(0.295)
$\Delta\ln\left(1+AVE_i^{Up}\right)$	-18.46***	-9.681	-11.08	-15.86*	-4.379***	-3.233	-0.310
	(6.503)	(7.312)	(7.326)	(8.269)	(1.249)	(6.743)	(0.596)
$\Delta\ln(p^*_{i,t-1})$			-0.271**	-0.243*	0.00490	-0.153	-0.0213
			(0.126)	(0.124)	(0.0155)	(0.124)	(0.0255)
$STS_i * \frac{AVE_i}{1+AVE_i}$			-1.995	-1.102	-0.799***	1.706	-0.0731
			(1.771)	(1.986)	(0.195)	(1.462)	(0.0966)
$I(MFA_i)$				-0.119	-0.00322	-0.106	-0.0136*
				(0.0947)	(0.0149)	(0.0742)	(0.00729)
$\Delta AVE_i{}^{Exports}$				-0.279	0.0210	0.0722	0.00269
				(0.255)	(0.0145)	(0.0912)	(0.0101)
Constant	1.197***	1.014***	1.038***	1.004***	0.0638***	0.462***	0.0113
	(0.113)	(0.0847)	(0.0868)	(0.104)	(0.00972)	(0.0637)	(0.00736)
IQE	0.006	-0.219	-0.232	-0.205	-0.031	-0.093	-0.009
IQR^{up}	-0.542	-0.055	-0.063	-0.090	-0.025	-0.018	-0.002
Estimator	OLS	IV	IV	IV	IV	IV	IV
Obs.	392	392	392	392	392	392	392
1st Stage Coeff.	.	0.000	0.000	0.000	0.000	0.000	0.000
KP F-Stat	.	74.737	72.269	53.941	53.941	53.941	53.941

Notes: Table presents estimates of equation 1 where the dependent variable is the change in log imports in columns 1-4, change in import penetration, which is imports divided by shipments minus exports plus imports, in column 5, change in log exports in column 5, and change in exports scaled by shipments in column 7. This table also includes upstream tariff changes similarly instrumented. AVE tariffs have been constructed at the SIC87 level with data from Greenland et al. (2025) and are instrumented in columns 2-7 with the Swiss formula implied tariff change as in equations 3 and 4. All data measured at or concorded to the 4-digit 1987 SIC classification. First stage coefficients and Kleibergen-Paap Rk F-statistic are reported in the footer of the table. IQE indicates the effect on the dependent variable an interquartile increase in $\Delta\ln(1+AVE_i)$ or $\Delta\ln(1+AVE_i^{Up})$ respectively. Robust standard errors reported in parentheses. *, **, and *** indicate p-values less than 10%, 5% and 1%, respectively.

Table A.6: Decomposing Pay Inequality

	(1)	(2)	(3)	(4)
	Panel (a): $\Delta \ln(\frac{Pay_i^{Non-Prod}}{Pay_i^{Prod}})$			
$\Delta \ln(1 + AVE_i)$	-1.804***	-2.994***	-2.839***	-2.705***
	(0.691)	(0.824)	(1.012)	(0.982)
IQE	-0.053	-0.088	-0.084	-0.080
	Panel (b): $\Delta \ln(\frac{Emp_i^{Non-Prod}}{Emp_i^{Prod}})$			
$\Delta \ln(1 + AVE_i)$	-1.495**	-2.432***	-1.569	-1.460
	(0.690)	(0.861)	(1.021)	(0.997)
IQE	-0.044	-0.072	-0.046	-0.043
	Panel (c): $\Delta \ln(\frac{Wage_i^{Non-Prod}}{Wage_i^{Prod}})$			
$\Delta \ln(1 + AVE_i)$	-0.309	-0.562	-1.270*	-1.245*
	(0.491)	(0.624)	(0.737)	(0.732)
IQE	-0.009	-0.017	-0.037	-0.037
Estimator	IV	IV	IV	IV
Controls	None	+Trade	+Prod.	+Tech.
Obs.	395	395	395	395
1st Stage Coeff.	0.150	0.144	0.135	0.136
Under ID p-stat	0.000	0.000	0.000	0.000
KP F-Stat	180.332	119.386	93.062	90.845

Notes: Table presents estimates of equation 1 where the dependent variable is the change in log pay, employment, and wages to non-production workers relative to production workers from 1979-1988 in panels a, b, and c respectively. AVE tariffs have been constructed at the SIC87 level with data from Greenland et al. (2025) and are instrumented with the Swiss formula implied tariff change as in equations 3 and 4. All data measured at or concorded to the 4-digit 1987 SIC classification. New controls are added sequentially and indicated in the footer. First stage coefficients, under-identified p-value and Kleibergen-Paap Rk F-statistic are reported in the footer of the table. IQE indicates the effect on the dependent variable of an interquartile increase in $\Delta \ln(1+AVE_i)$. Robust standard errors reported in parentheses. *, **, and *** indicate p-values less than 10%, 5% and 1%, respectively.

Table A.7: Tariff Liberalization and Pay Inequality – Alternate IV

	$\Delta \ln(\frac{Pay_i^{Non-Prod}}{Pay_i^{Prod}})$			
	(1)	(2)	(3)	(4)
$\Delta \ln(1 + AVE_i)$	-1.771**	-2.952***	-2.764***	-2.598***
	(0.693)	(0.826)	(1.011)	(0.981)
IQR	-0.216	-0.360	-0.337	-0.317
1st Stage Coeff.	-0.093	-0.090	-0.085	-0.085
Under ID p-stat	0.000	0.000	0.000	0.000
KP F-Stat	171.911	113.729	88.740	86.816
Estimator	IV	IV	IV	IV
Controls	None	+ Trade	+ Prod.	+Tech

Notes: Table presents estimates of equation 1 where the dependent variable is the change in log pay inequality. AVE tariffs have been constructed at the 4-digit 1987 SIC industry level with data from Greenland et al. (2025) and are instrumented with the initial column 2 tariff level. All data measured at or concorded to the 4-digit 1987 SIC classification. First stage coefficients, under-identified p-value and Kleibergen-Paap Rk F-statistic are reported in the footer of the table. Robust standard errors reported in parentheses. *, **, and *** indicate p-values less than 10%, 5% and 1%, respectively.

Table A.8: $\Delta \ln \left(\frac{Pay_{Non-Prod}}{Pay_{Prod}} \right)$

	Panel (a): All Workers			
$\Delta \ln (1 + AVE_i)$	-2.729**	-4.163***	-2.940*	-3.270*
	(1.180)	(1.526)	(1.690)	(1.693)
IQE	-0.075	-0.115	-0.081	-0.090
	Panel (b): Male Workers			
$\Delta \ln (1 + AVE_i)$	-1.491	-2.509	-2.072	-2.475
	(1.181)	(1.678)	(1.997)	(1.990)
IQE	-0.041	-0.069	-0.057	-0.068
	Panel (c): Female Workers			
$\Delta \ln (1 + AVE_i)$	-9.410***	-10.55***	-7.175***	-6.783***
	(1.937)	(2.228)	(2.419)	(2.093)
IQE	-0.260	-0.292	-0.198	-0.187
Estimator	IV	IV	IV	IV
Controls	Full	Full	Full	Full
Obs.	72	72	72	72
1st Stage Coeff.	0.138	0.136	0.124	0.130
Under ID p-stat	0.000	0.000	0.000	0.000
KP F-Stat	63.995	40.873	32.265	32.136

Notes: Table presents estimates of equation 1 where the dependent variable differs based on the panel. AVE tariffs have been constructed at the 4-digit SIC87 industry level with data from Greenland et al. (2025) and are instrumented with the Swiss formula implied tariff change as in 4. All data measured at or concorded to the 4-digit 1987 SIC classification. First stage coefficients, under-identified p-value and Kleibergen-Paap Rk F-statistic are reported in the footer of the table. Robust standard errors reported in parentheses. *, **, and *** indicate p-values less than 10%, 5% and 1%, respectively.

Table A.9: Pretrends for Effects of Tariff Liberalization for Men and Women

	All	Men	Women
	Panel (a): $\Delta \ln(\frac{Pay_i^{Non-Prod}}{Pay_i^{Prod}})_{t-1}$		
$\Delta \ln(1 + AVE_i)$	2.282	1.565	3.792
	(3.521)	(3.755)	(3.850)
IQR	0.063	0.043	0.105
Estimator	IV	IV	IV
Controls	Full	Full	Full
Obs.	72	72	72
1st Stage Coeff.	0.130	0.130	0.130
Under ID p-stat	0.000	0.000	0.000
KP F-Stat	32.136	32.136	32.136
	Panel (b): $\Delta \ln(\frac{Pay_i^{Non-Prod}}{Pay_i^{Prod}})$		
$\Delta \ln(1 + AVE_i)$	-2.921*	-2.279	-6.998***
	(1.630)	(1.953)	(2.041)
$\Delta \ln(\frac{Pay_i^{Non-Prod,A}}{Pay_i^{Prod,A}})_{t-1}$	-0.153**		
	(0.0667)		
$\Delta \ln(\frac{Pay_i^{Non-Prod,M}}{Pay_i^{Prod,M}})_{t-1}$		-0.125*	
		(0.0681)	
$\Delta \ln(\frac{Pay_i^{Non-Prod,W}}{Pay_i^{Prod,W}})_{t-1}$			0.0568
			(0.0748)
IQR	-0.081	-0.063	-0.193
Estimator	IV	IV	IV
Controls	Full	Full	Full
Obs.	72	72	72
1st Stage Coeff.	0.130	0.130	0.134
Under ID p-stat	0.000	0.000	0.000
KP F-Stat	31.692	31.880	33.652

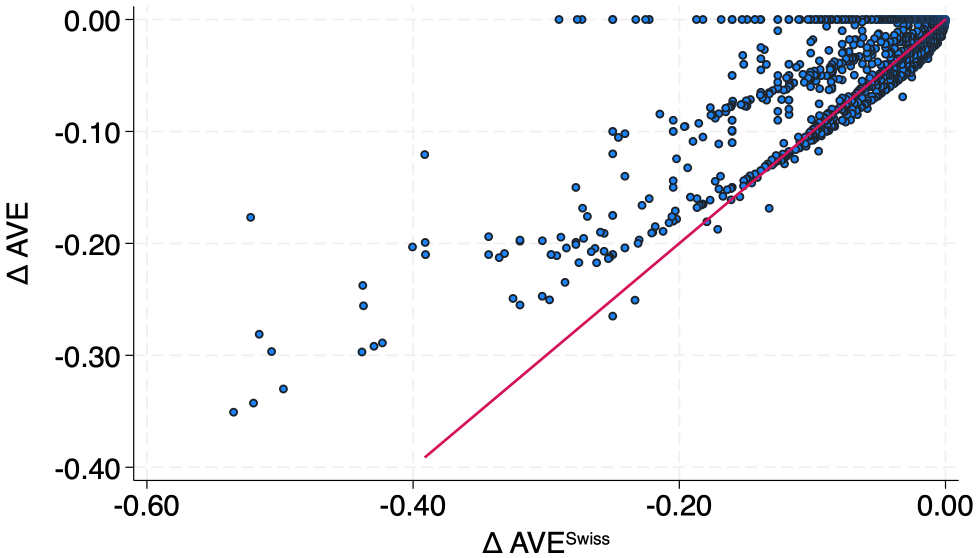
Notes: Table presents estimates of equation 1 where the dependent variable differs based on the panel. Dependent variable is change in log pay inequality from 1970-1980 as a placebo exercise and controls for lagged change in log pay inequality in panel B. Column 1 reports estimates for all workers, column 2 reports estimates for men, and column 3 reports estimates for women. AVE tariffs have been constructed at the 4-digit SIC87 industry level with data from Greenland et al. (2025) and are instrumented with the Swiss formula implied tariff change as in 4. All data measured at or concorded to the 4-digit 1987 SIC classification. First stage coefficients, under-identified p-value and Kleibergen-Paap Rk F-statistic are reported in the footer of the table. Robust standard errors reported in parentheses. *, **, and *** indicate p-values less than 10%, 5% and 1%, respectively.

Table A.10: Tariff Liberalization and College Premium

	$\Delta \ln(\frac{Pay_i^{College}}{Pay_i^{Non-College}})$		
	All	Men	Women
$\Delta \ln(1 + AVE_i)$	-3.270*	-1.350	-7.060*
	(1.693)	(2.269)	(4.034)
IQE	-0.090	-0.037	-0.195
Specification	IV	IV	IV
Obs.	72	72	72
1st Stage Coeff.	0.130	0.130	0.130
Under ID p-stat	0.000	0.000	0.000
KP F-Stat	32.136	32.136	32.136

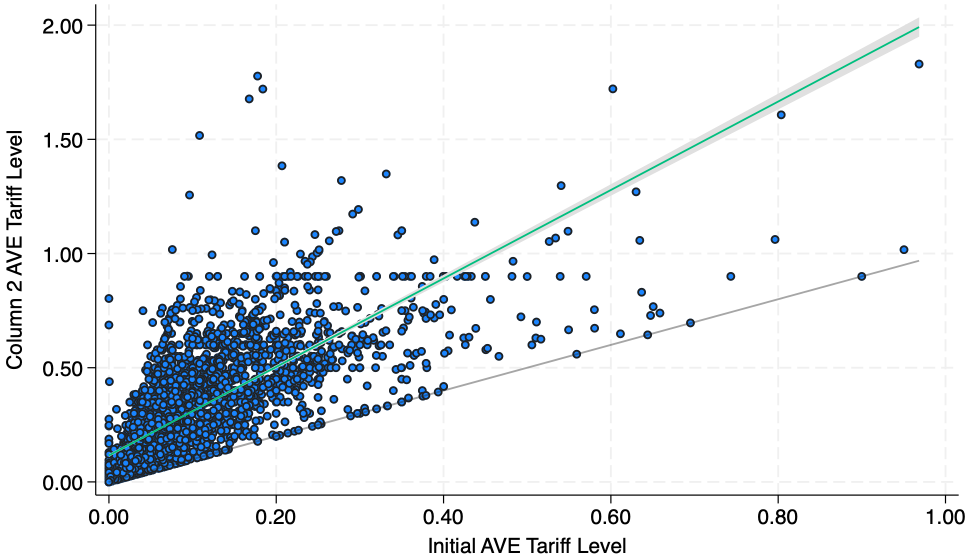
Notes: Table presents estimates of equation 1 where the dependent variable is change in log pay inequality from 1980-1990 among those with a college education as opposed to those without. AVE tariffs have been constructed at the 4-digit SIC87 industry level with data from Greenland et al. (2025) and are instrumented with the Swiss formula implied tariff change as in 4. All data measured at or concorded to the census industry classification. Data from Ruggles et al. (2024). First stage coefficients, under-identified p-value and Kleibergen-Paap Rk F-statistic are reported in the footer of the table. Robust standard errors reported in parentheses. *, **, and *** indicate p-values less than 10%, 5% and 1%, respectively.

Figure A.1: AVE Tariff Cuts During the Tokyo Round



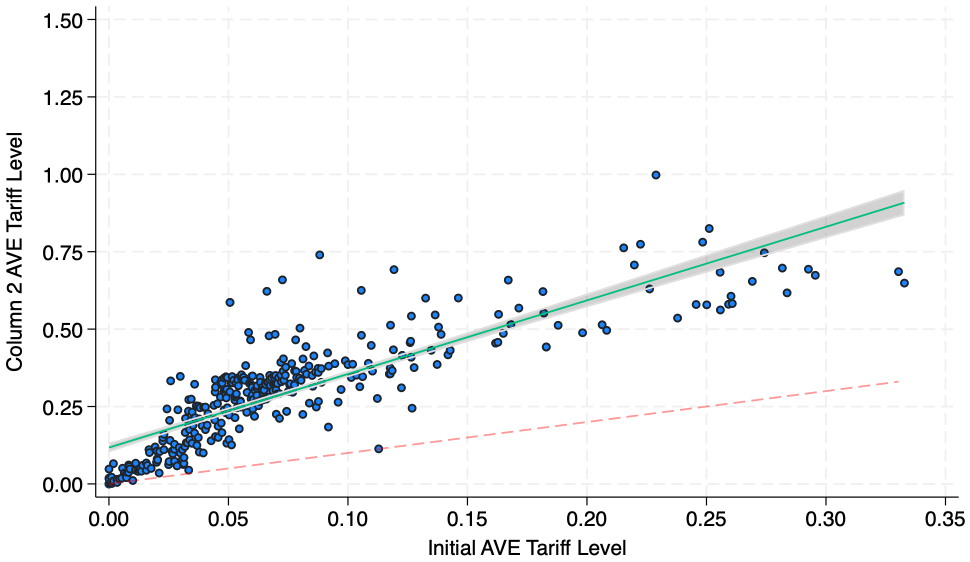
Notes: Figure displays observed (vertical axis) versus tariff cut implied by the Swiss formula (horizontal axis) at the 5-digit TSUSA level between 1979 and 1988 as described in equation 2. Strict adherence to the Swiss formula indicated by the red line. Sample restricted to set of codes observed in both years. Source: Greenland et al. (2025) and authors' calculations.

Figure A.2: Observed versus Column 2 Industry-level AVE Tariffs, TSUSA 5-digit



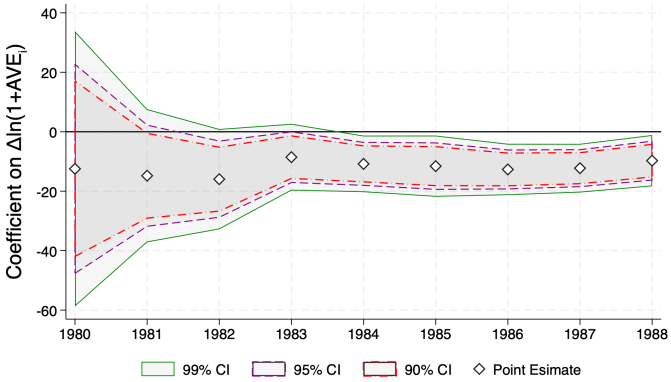
Notes: The figure depicts observed 1979 AVE tariffs versus their column 2 counterpart at the 5-digit TSUSA level. Data from Greenland et al. (2025) and authors' calculations.

Figure A.3: Observed versus Column 2 Industry-level AVE Tariffs, SIC87 4-digit

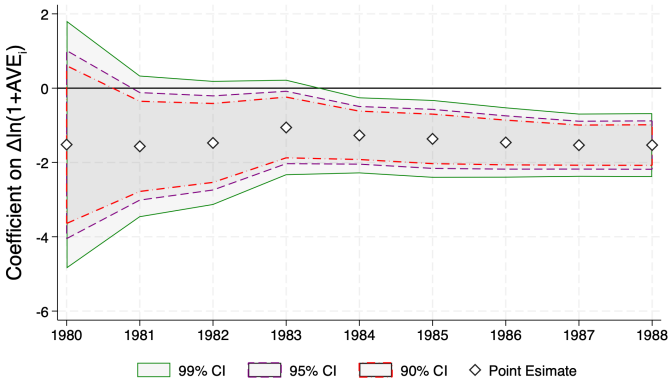


Notes: The figure depicts observed 1979 AVE tariffs and 1978 column 2 AVE tariffs for the 392 4-digit 1987 SIC industries in our sample. Correlation is 82.9%. Data from Greenland et al. (2025) and authors' calculations.

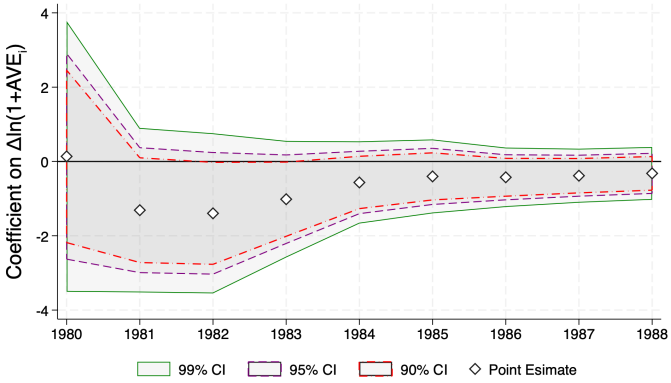
Figure A.4: Changes in Trade Exposure Measures Over Time



(a) $\Delta \ln(Imports)$



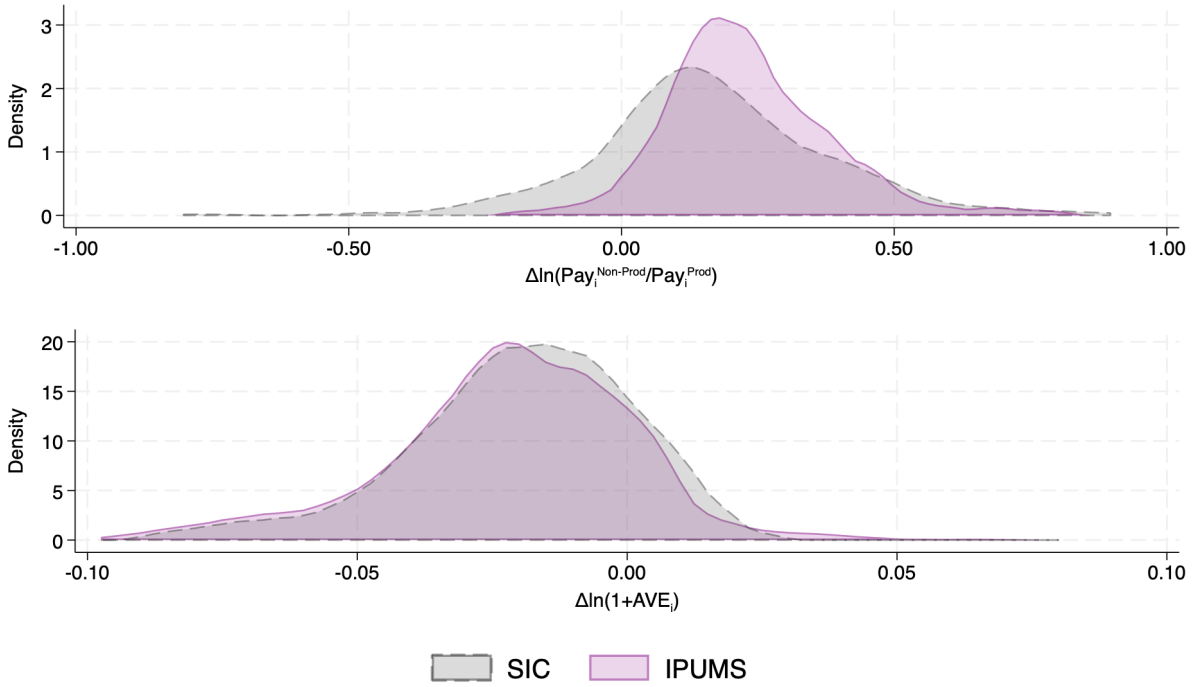
(b) ΔImport Penetration



(c) $\Delta \frac{Exports_{it}}{Shipments_{it}}$

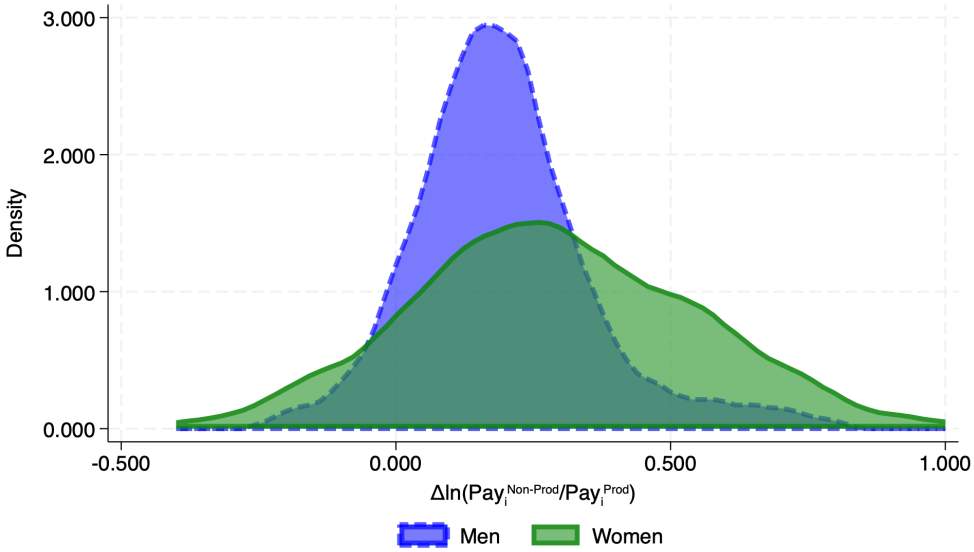
Notes: Figure presents estimates of equation 1 where the dependent variable is the change in log imports, change in import penetration, and change in exports divided by shipments. Each point is a coefficient from a different regression using a time horizon from 1979-t where t is between 1980 and 1988. AVE tariffs from Greenland et al. (2025) and are instrumented with the Swiss formula implied tariff change as in equations 3 and 4. Outcome data taken from the NBER manufacturing database. Data measured at or concorded to the 4-digit SIC87 industry classification. Confidence intervals reflect robust standard errors

Figure A.5: Distribution of Pay Inequality and Tariff Changes by Dataset



Notes: Figure presents distribution of growth in log pay inequality and change in observed tariff rates ($\Delta \ln(1 + AVE_i)$ when aggregated to the 395 4-digit SIC87 industries and the 72 census industries for the IPUMS (Ruggles et al., 2024) analysis.

Figure A.6: Growth in Pay Inequality by Sex



Notes: Figure presents distribution of growth in log pay inequality for men and women when aggregated to the 72 manufacturing census industries in our analysis. Data from IPUMS (Ruggles et al., 2024) and author's calculations.

B Data Appendix

B.1 Aggregating Legislated Tariffs to SIC Industry Tariffs

Our goal is to calculate annual AVE tariffs at the 4-digit SIC level, defined as duties collected divided by the value of imports. To do so, we concord from the disaggregate TSUSA level to the 4-digit SITC rev. 2 classification and then to the 4-digit SIC 1987 system via the 6-digit HS level. Two issues emerge immediately. First, the concordance from the TSUSA system to the 4-digit SITC rev. 2 level maps uniquely from 7-digit TSUSA codes even though legislated tariffs are defined at the 5-digit TSUSA level. Second, exporters of a given 7-digit TSUSA good may face one of three tariffs: (i) the vast majority of exporters and products face the column 1 tariffs but, in practice during our sample period, (ii) a small subset of communist countries face column 2 tariffs, and (iii) a subset of products from a subset of developing countries have duty free access (e.g. due to the US Generalized System of Preferences (GSP) program). To this end, we use annual legislated tariffs between 1978 and 1988 from Greenland et al. (2025) at the exporter by 7-digit level. We also take import data provided by Greenland et al. (2025) at the same level of disaggregation.

Starting at the 7-digit TSUSA level, duties collected on good k in year t across exporters n are

$$d_{kt} = \sum_{n} \left(\tau_{knt} m_{knt} + f_{knt} q_{knt} \right) \tag{B.1}$$

where τ is the ad valorem tariff, f is the specific tariff, q is the quantity of imports, and m is the value of imports. Aggregating across the set of 7-digit TSUSA goods $K(j)$ that uniquely map to 4-digit SITC j gives

$$d_{jt} = \sum_{k \in K(j)} d_{kt} \tag{B.2}$$

$$m_{jt} = \sum_{k \in K(j)} m_{kt}. \tag{B.3}$$

For later purposes, it is useful to note that we can always rewrite an aggregate AVE tariff – defined as aggregate duties collected divided by the aggregate value of imports (for example, aggregated across 7-digit TSUSA goods k to the 4-digit SITC industry j) – as an import weighted average of disaggregate AVE tariffs. To this end, let $Col1$ and $Col2$ denote that

imports are subject to column 1 and column 2 tariffs. Then,

$$
\begin{aligned}
AVE_{jt} \equiv \frac{d_{jt}}{m_{jt}} &= \left[\sum_{k\in K(j)} d_{kt}^{Col1} + \sum_{k\in K(j)} d_{kt}^{Col2}\right]\left[\sum_{k\in K(j)} m_{kt}\right]^{-1} \\
&= \frac{1}{m_{jt}} \sum_{k\in K(j)} \left(\frac{d_{kt}^{Col1}}{m_{kt}^{Col1}} m_{kt}^{Col1} + \frac{d_{kt}^{Col2}}{m_{kt}^{Col2}} m_{kt}^{Col2}\right) \\
&\equiv \frac{1}{m_{jt}} \sum_{k\in K(j)} \left(AVE_{kt}^{Col1} m_{kt}^{Col1} + AVE_{kt}^{Col2} m_{kt}^{Col2}\right) \\
&= \sum_{k\in K(j)} \left(AVE_{kt}^{Col1} \frac{m_{kt}^{Col1}}{m_{kt}} \frac{m_{kt}}{m_{jt}} + AVE_{kt}^{Col2} \frac{m_{kt}^{Col2}}{m_{kt}} \frac{m_{kt}}{m_{jt}}\right) \\
&\equiv \sum_{k\in K(j)} \left(AVE_{kt}^{Col1} \omega_{kt}^{Col1} + AVE_{kt}^{Col2} \omega_{kt}^{Col2}\right) \omega_{kjt} \qquad \text{(B.4)} \\
&\equiv \sum_{k\in K(j)} AVE_{kt} \omega_{kjt}. \qquad \text{(B.5)}
\end{aligned}
$$

Equation (B.5) says the aggregate AVE tariff AVE_{jt} is simply the import weighted average of the disaggregate AVE tariffs AVE_{kt} with the weights ω_{kjt} denoting the time-varying share of industry j imports accounted for by good k. And, equation (B.4) says each disaggregate AVE tariff AVE_{kt} is the import weighted average AVE tariff across column 1 imports, AVE_{kt}^{Col1}, and column 2 imports, AVE_{kt}^{Col2}, with the weights ω_{kt}^{Col1} and ω_{kt}^{Col2} denoting, respectively, the time-varying share of good k imports accounted for by column 1 and column 2 imports.[43]

To concord from 4-digit SITC industries j to 4-digit SIC industries i, we use the HS system as an intermediate step. In general, 4-digit SITC codes are more aggregate than 6-digit HS codes. Thus, a 4-digit SITC code j typically maps to multiple 6-digit HS codes. Denote the number of such 6-digit HS codes by l_j^{HS}. Similarly, 6-digit HS6 codes are generally more disaggregate than 4-digit SIC codes. Thus, a given 6-digit HS code h typically maps to a unique 4-digit SIC code. Nevertheless, denote the number of such 4-digit SIC codes by l_h^{SIC}. Then the number of "connections" from 4-digit SITC code j to 4-digit SIC code i is $l_j^{HS} l_h^{SIC}$. Naturally, some of these connections start with the same 4-digit SITC code j and end with the same 4-digit SIC code i. Thus, denote the number of such unique 4-digit SIC codes i "connected" from 4-digit SITC code j by l_j^{SIC}.[44]

[43] Equation (B.4) omits a third term for the category of imports subject to zero tariffs due to, e.g. the US GSP program. This is without loss of generality because these imports have zero AVE tariffs and still enter the total import terms m_{kt} and m_{jt}.

[44] For example, SITC code 0011 maps to the two HS codes 010210 and 010290 (i.e. $l_j^{HS} = 2$). And, each

Let $J(i)$ denote the set of SITC industries j that concord to SIC industry i. Then, the AVE tariff for 4-digit SIC industry i is the weighted sum of duties collected across 4-digit SITC industries $j \in J(i)$ divided by the weighted sum of imports across these 4-digit SITC industries:

$$\begin{aligned} AVE_{it} &= \left[\sum_{j\in J(i)} \frac{1}{l_j^{SIC}} d_{jt}\right] \left[\sum_{j\in J(i)} \frac{1}{l_j^{SIC}} m_{jt}\right]^{-1} \\ &\equiv \left[\sum_{j\in J(i)} d_{jit}\right] \left[\sum_{j\in J(i)} m_{jit}\right]^{-1} \\ &= \sum_{j\in J(i)} AVE_{jt}\omega_{jit} \end{aligned} \tag{B.6}$$

where the last line follows from the general interpretation of an aggregate AVE in equation (B.5). Equation (B.6) says the AVE tariff for 4-digit SIC industry i, AVE_{it}, is the import-weighted average of AVEs across the associated 4-digit SITC industries j, AVE_{jt}, where the weights ω_{jit} correspond to the time varying share of industry i's imports accounted for by industry j. And, as described above, equation (B.5) says the AVE tariff for SITC industry industry j is the import-weighted average AVE tariff across 7-digit TSUSA goods k which, per equation (B.4), is the import weighted average AVE tariff across good k column 1 and column 2 imports. Thus, ultimately, the AVE tariff for a 4-digit SIC industry i is an import weighted average of the AVE of the underlying legislated tariffs at the TSUSA tariff-line level.

Construction of our instruments now follows naturally in two steps. First, we calculate an instrumented value for AVE_{kt} with fixed 1978 import weights to substitute into equation (B.5). This gives instrumented values for AVE_{jt}. Second, we substitute the instrumented values for AVE_{jt} into equation (B.6) using fixed 1978 import weights. This gives instrumented values for AVE_{it}.

Our Swiss IV substitutes column 2 tariffs into the 1978 Swiss formula and phases out linearly between 1980 and 1987. Note that Figure A.1 in Appendix **??** re-creates Figure 4 at the tariff line 5-digit TSUSA level. And, Figure A.1 in Appendix **??** plots the column 1 1978 AVE tariff against the 1978 column 2 AVE tariff at the tariff-line 5-digit TSUSA level.

of these HS codes map to the two SIC codes 0211 and 0241 (i.e. $l_h^{SIC} = 2$ for each of the two HS codes). But, among these $l_j^{HS} l_h^{SIC} = 4$ connections, the SITC code 0011 only maps to two SIC codes (i.e. l_j^{SIC}=2): 0211 and 0241.

In 1979, the instrument value for good g at the 5-digit TSUSA level is

$$\begin{aligned} AVE^{IV}_{g,1979} &= \frac{\sum_n \left(\tau^{Col2}_{gn,1978} m^{Col1}_{gn,1978} + f^{Col2}_{gn,1978} q^{Col1}_{gn,1978}\right)}{m^{Col1}_{g,1978}} \\ &= \tau^{Col2}_{g,1978} + \frac{f^{Col2}_{g,1978}}{p_{g,1978}} \end{aligned} \tag{B.7}$$

where $p_{g,1978}$ is the 1978 unit value for 5-digit TSUSA good g column 1 imports. Once the US has fully implemented their Tokyo tariff cuts in $t = 1987, 1988$, the instrument value is:

$$AVE^{IV}_{gt} = \frac{0.14 \times AVE^{IV}_{g,1979}}{0.14 + AVE^{IV}_{g,1979}} \text{ for } t = 1987, 1988 \tag{B.8}$$

For the phase-out years $t = 1980, ..., 1986$, we phase out AVE^{IV}_{gt} linearly between $AVE^{IV}_{g,1979}$ and $AVE^{IV}_{g,1987}$. Substituting AVE^{IV}_{gt} for AVE^{col1}_{gt} in equation (B.4) and fixing import weights at their 1978 values gives:

$$\begin{aligned} AVE^{IV}_{jt} &\equiv \sum_{k \in K(j)} \left(AVE^{IV}_{g(k),t} \omega^{Col1}_{k,1978} + AVE^{Col2}_{g(k),1978} \omega^{Col2}_{k,1978}\right) \omega_{kj,1978} \\ &= \sum_{g \in G(j)} \left(AVE^{IV}_{gt} \omega^{Col1}_{g,1978} + AVE^{Col2}_{g,1978} \omega^{Col2}_{g,1978}\right) \omega_{gj,1978} \\ &\equiv \sum_{g \in G(j)} AVE^{IV}_{gt} \omega_{gj,1978}. \end{aligned} \tag{B.9}$$

where $g(k)$ in the first line denotes the 5-digit TSUSA good g associated with 7-digit TSUSA good k. Finally, substituting equation (B.9) into equation (B.6) gives

$$\begin{aligned} AVE^{IV}_{it} &\equiv \sum_{j \in J(i)} AVE^{IV}_{jt} \omega_{ji,1978} \\ &= \sum_{j \in J(i)} \sum_{g \in G(j)} AVE^{IV}_{gt} \omega_{gj,1978} \omega_{ji,1978}. \end{aligned} \tag{B.10}$$

That is, the value of our Swiss IV for 4-digit SIC industry i is an import weighted average of the Swiss IV at the tariff line 5-digit TSUSA level g.

B.2 Exclusion restriction

Our exclusion restriction when estimating equation (1) using our Swiss IV is

$$\begin{aligned}
\operatorname{cov}\left(\frac{zAVE_{it}^{IV}}{z+AVE_{it}^{IV}},\varepsilon_{it}\right)&=0\\
\mathrm{E}\left[\left(\frac{zAVE_{it}^{IV}}{z+AVE_{it}^{IV}}-\mathrm{E}\left(\frac{zAVE_{it}^{IV}}{z+AVE_{it}^{IV}}\right)\right)\varepsilon_{it}\right]&=0 \text{ using } \mathrm{E}\left(u\right)=0\\
\mathrm{E}\left[\frac{zAVE_{it}^{IV}}{z+AVE_{it}^{IV}}\varepsilon_{it}-\mathrm{E}\left(\frac{zAVE_{it}^{IV}}{z+AVE_{it}^{IV}}\right)\varepsilon_{it}\right]&=0\\
\mathrm{E}\left(\frac{zAVE_{it}^{IV}}{z+AVE_{it}^{IV}}\varepsilon_{it}\right)-\mathrm{E}\left(\frac{zAVE_{it}^{IV}}{z+AVE_{it}^{IV}}\right)\mathrm{E}\left(\varepsilon_{it}\right)&=0\\
\mathrm{E}\left(\frac{zAVE_{it}^{IV}}{z+AVE_{it}^{IV}}\varepsilon_{it}\right)&=0 \text{ using } \mathrm{E}\left(u\right)=0\\
\mathrm{E}\left(\frac{zAVE_{it}^{IV}\varepsilon_{it}}{z+AVE_{it}^{IV}}\right)&=0. \qquad \text{(B.11)}
\end{aligned}$$

Using the standard Taylor approximation for the expectation of a non-linear function of random variables, we have

$$\mathrm{E}\left(\frac{zAVE_{it}^{IV}\varepsilon_{it}}{z+AVE_{it}^{IV}}\right)\approx\frac{\mathrm{E}\left(zAVE_{it}^{IV}\varepsilon_{it}\right)}{\mathrm{E}\left(z+AVE_{it}^{IV}\right)}-\frac{\operatorname{cov}\left(zAVE_{it}^{IV}\varepsilon_{it},AVE_{it}^{IV}\right)}{\left[\mathrm{E}\left(z+AVE_{it}^{IV}\right)\right]^{2}}+\frac{\operatorname{var}\left(z+AVE_{it}^{IV}\right)\mathrm{E}\left(zAVE_{it}^{IV}\varepsilon_{it}\right)}{\left[\mathrm{E}\left(z+AVE_{it}^{IV}\right)\right]^{3}}=0 \qquad \text{(B.12)}$$

Assuming exogeneity of the column 2 tariffs says that

$$\mathrm{E}\left(AVE_{it}^{IV}\varepsilon_{it}\right)=0 \qquad \text{(B.13)}$$

and hence, given $z>0$ does not vary across i or t, equation (B.12) reduces to

$$\begin{aligned}
\mathrm{E}\left(\frac{zAVE_{it}^{IV}\varepsilon_{it}}{z+AVE_{it}^{IV}}\right)\approx&-\frac{\operatorname{cov}\left(zAVE_{it}^{IV}\varepsilon_{it},AVE_{it}^{IV}\right)}{\left[\mathrm{E}\left(z+AVE_{it}^{IV}\right)\right]^{2}}=0\\
\Leftrightarrow&\operatorname{cov}\left(zAVE_{it}^{IV}\varepsilon_{it},AVE_{it}^{IV}\right)=0\\
\Leftrightarrow&z\operatorname{cov}\left(AVE_{it}^{IV}\varepsilon_{it},AVE_{it}^{IV}\right)=0\\
\Leftrightarrow&\operatorname{cov}\left(AVE_{it}^{IV}\varepsilon_{it},AVE_{it}^{IV}\right)=0. \qquad \text{(B.14)}
\end{aligned}$$

Finally, using $\mathrm{E}(u) = 0$ and equation (B.13), our exclusion restriction restriction from equation (B.14) can be expressed as

$$\begin{aligned}
\operatorname{cov}\left(AVE_{it}^{IV}\varepsilon_{it}, AVE_{it}^{IV}\right) &= \mathrm{E}\left[\left(AVE_{it}^{IV}\varepsilon_{it} - \mathrm{E}\left(AVE_{it}^{IV}\varepsilon_{it}\right)\right)\left(AVE_{it}^{IV} - \mathrm{E}AVE_{it}^{IV}\right)\right] \\
&= \mathrm{E}\left[AVE_{it}^{IV}\varepsilon_{it}\left(AVE_{it}^{IV} - \mathrm{E}AVE_{it}^{IV}\right)\right] \\
&= \mathrm{E}\left[\left(AVE_{it}^{IV}\right)^2\varepsilon_{it} - AVE_{it}^{IV}\varepsilon_{it}\mathrm{E}AVE_{it}^{IV}\right] \\
&= \mathrm{E}\left[\left(AVE_{it}^{IV}\right)^2\varepsilon_{it}\right] - \mathrm{E}\left[AVE_{it}^{IV}\varepsilon_{it}\right]\mathrm{E}AVE_{it}^{IV} \\
&= \mathrm{E}\left[\left(AVE_{it}^{IV}\right)^2\varepsilon_{it}\right] = 0 \qquad \text{(B.15)} \\
&\Leftrightarrow \operatorname{cov}\left[\left(AVE_{it}^{IV}\right)^2, \varepsilon_{it}\right] = 0 \qquad \text{(B.16)}
\end{aligned}$$

This says that, in addition to the exogeneity of the column 2 tariff as expressed in equation (B.13), we also need the square of the column 2 tariff to be uncorrelated with unobservables ε_{it}.

B.3 Upstream Input Tariffs

Our measure of input tariffs faced by industry i is a weighted average of AVE tariffs on all manufacturing inputs z used by industry i. The time-invariant weights depend on (i) s_{iz}, the total requirement for industry i of input z per dollar of industry i output from the 1977 IO table from the US Bureau of Economic Analysis, and (ii) 1978 US imports of z, m_z, as a share of 1978 US manufacturing imports (Schott, 2008). Specifically,

$$AVE_{it}^{Up} = \sum_z AVE_{zt}\omega_z \tag{B.17}$$

where the time-invariant weight is

$$\omega_z = \frac{s_{iz}\frac{m_z}{\sum_z m_z}}{\sum_z s_{iz}\frac{m_z}{\sum_z m_z}}. \tag{B.18}$$

Finally, using equation (B.10) to substitute AVE_{zt}^{IV} into (B.17) for AVE_{zt} gives our instrument for AVE_{it}^{Up}. Note that, for industry i, the weight on its input tariff z relative to the weight on its input tariff z' is

$$\frac{\omega_z}{\omega_{z'}} = \frac{s_{iz}}{s_{iz'}}\frac{m_z}{m_{z'}}. \tag{B.19}$$

Intuitively, this relative weight is higher when (i) industry i uses more of input z relative to input z' and/or (ii) the US imports more of input z than input z'.

B.4 Tariff Changes by Foreign Countries

We have digitized data on the tariff reductions negotiated by Japan and the European Economic Community (EEC), taken from the WTO website: https://www.wto.org/english/docs_e/gattbilaterals_e/indexbyround_e.htm.

We have chosen these countries because they are the primary export destinations for US manufacturing, covering over half of the value of all US exports in 1979. We display the share of US manufacturing exports destined for a given location as of 1979 in Figure B.1. Other than Mexico, no remaining country has an export share above 3%, and while ROW exports accounts for roughly 17% of US manufacturing export value, this is divided among 129 destinations.

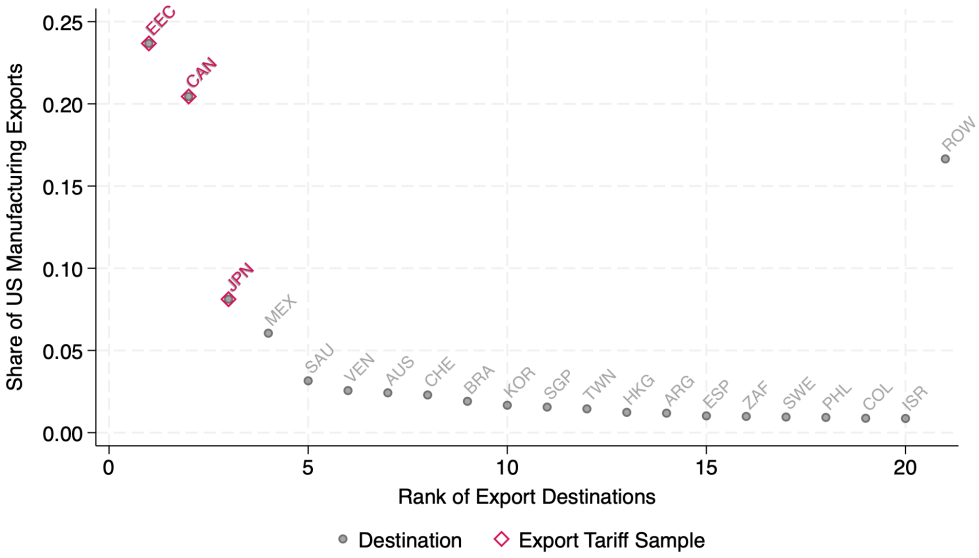
Figure B.1 about here.

We manually concord tariff lines to the Customs Co-operation Council Nomenclature (CCCN) and map these to 1992 HS codes using a concordance provided by WITS: https://wits.worldbank.org/product_concordance.html. Finally, we concord these to the SIC using the concordance created by Pierce and Schott (2012). We also take Canadian import tariffs from the Canadian SIC classification and concord these to the US SIC classification using data and a concordance from Trefler (2004). Finally, we weight these SIC-level tariff changes in each member by the share of 1972 US exports among the three partners accounted for by each market.

The WTO/GATT tariff files only provide information on tariffs which have been changed from their initial value. Consequently we cannot construct $\Delta \ln(1 + AVE_i^{Exports})$ as requires data on initial tariff levels prior to aggregation. As a result we use $\Delta AVE_i^{Exports}$ which does not depend on initial tariff levels. This measure is correlated with US tariff cuts, as can be seen in Figure B.2.

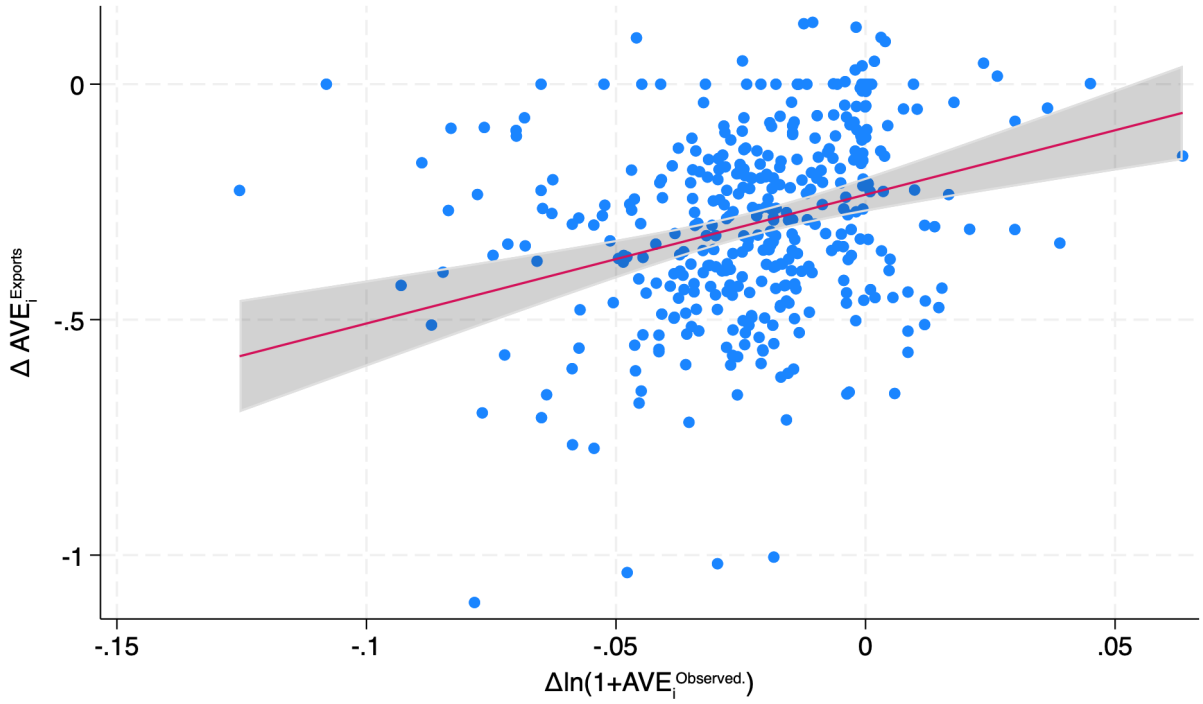
Figure B.2 about here.

Figure B.1: Destination's Share of US Exports in 1979



Notes: Figure displays export shares of the 21 destinations for US exports in 1979. Data from Schott (2008)

Figure B.2: Correlation Between Import and Export Liberalization during the Tokyo Round of the GATT



Notes: Figure presents the correlation between cuts in US import tariffs and export tariffs among the countries highlighted in Figure B.1. One observation (driven by a large increase in tariffs related to malt and brewing in Canada) has been omitted for readability. Data for EEC and Japan have been digitized from https://www.wto.org/english/docs_e/gattbilaterals_e/indexbyround_e.htm. Canadian tariffs taken from Trefler (2004).

C Endogenous Liberalization

An appealing feature of our setting is that we are able to explore the nature of endogenous protection: what factors drove US policymakers to deviate from the Swiss formula? Specifically, we explore the characteristics of the under-liberalized industries relative to the over-liberalized industries. This is a rare opportunity, as studies analyzing the causal impact of trade policy typically require settings in which the observed policy changes are plausibly exogenous, ruling out an exploration of the endogenous determinants of those changes by definition, or require that all such determinants have been identified. By exploiting the planned tariff reductions from the Swiss formula, we are able to identify the causal effects of Tokyo Round tariff reductions while simultaneously identifying endogenous correlates of observed protection.

Figure C.1 about here.

Motivated by our focus on inequality between production and non-production workers, we begin by highlighting the imprint of industry skill intensity on the Tokyo Round tariff cuts chosen by policymakers. Figure C.1 presents separate kernel densities of the 1978 non-production labor share of employment within the 198 under-liberalized (blue) industries and the 197 over-liberalized industries (red). While there is considerable overlap in the densities, the overall pattern is stark: under-liberalized industries tend to have smaller employment shares of non-production labor than over-liberalized industries. That is, policymakers are more likely to deviate from the Swiss formula to protect industries with lower pre-Tokyo skill intensity.

We analyze this relationship more formally in Table C.1. Specifically, we estimate a linear probability model relating 1978 industry characteristics to whether or not an industry is under-liberalized. Column (1) of the table includes only log of one plus 1978 observed industry tariff levels. Column 2 introduces our trade controls including tariffs by foreign countries, lagged industry price growth, a dummy variable for having MFA exposure, and a control for the inflationary erosion of specific tariffs. Column 3 introduces capital to labor and women's employment shares. Column 4 introduces non-production employment shares. Finally, column 5 introduces our technology-related controls.

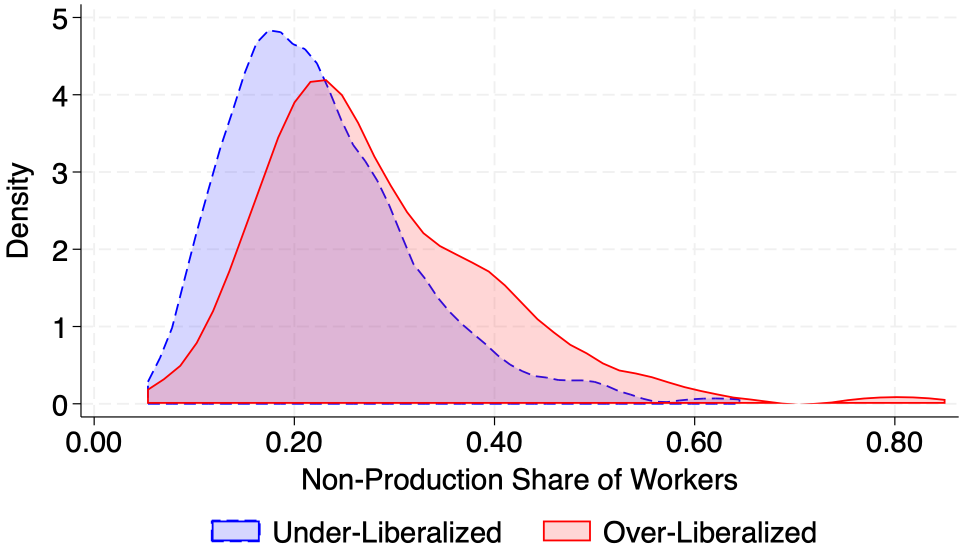
Three characteristics seem to predict under-liberalization. First, industries experiencing substantial lagged price growth are more likely to be under-liberalized (have their tariffs cut by less than prescribed by the Swiss formula). As documented in Greenland et al. (2025), the years leading up to the Tokyo round experienced rampant inflation which eroded the

protective capacity of specific tariffs. As such, many of the sectors experiencing industry price growth were liberalized during the 1970s. These industries seem to have been partially exempted from the full tariff cuts. Second, industries whose labor forces were tilted toward production workers were also differentially protected. This is consistent with our findings in Figure C.1. Lastly, industries with large concentrations of occupations heavily characterized by routine tasks less likely to have been exempted from the full tariff cuts implied by the Swiss formula.

Table C.1 about here.

These relationships suggests that politicians appear to have considered pre-existing characteristics in a way that materially impacted the distribution of Tokyo liberalization across industries. Naturally, such endogenous manipulation of tariff changes may yield biased estimates of the effects of observed changes on industry outcomes. This motivates the IV approach taken in the paper.

Figure C.1: Non-production Labor Shares and Endogenous Liberalization



Notes: Kernel density of non-production labor shares of employment displayed separately for 4-digit 1987 SIC industries receiving lower tariff cuts than prescribed by the Swiss formula (Under-Liberalized) versus those receiving greater tariff cuts than prescribed (Over-Liberalized). Data from Greenland et al. (2025), the NBER-CES manufacturing database, and authors' calculations.

Table C.1: Determinants of Under-Liberalization

	(1)	(2)	(3)	(4)	(5)
$\ln(1 + AVE_i)$	2.804***	1.135*	1.222	0.833	1.110
	(0.366)	(0.649)	(0.779)	(0.774)	(0.774)
$\Delta AVE_i{}^{Exports}$		0.007	0.000	0.019	-0.066
		(0.095)	(0.096)	(0.100)	(0.092)
$\Delta \ln(p^*_{i,t-1})$		0.445***	0.438***	0.411***	0.313**
		(0.136)	(0.135)	(0.130)	(0.130)
$I(MFA_i)$		0.223***	0.214***	0.161**	0.120
		(0.075)	(0.080)	(0.079)	(0.081)
$STS_i * \frac{AVE_i}{1+AVE_i}$		2.925**	2.440*	2.040	1.621
		(1.156)	(1.251)	(1.237)	(1.278)
$\frac{Capital_i}{Labor_i}$			0.038	0.047	0.024
			(0.039)	(0.039)	(0.045)
$\frac{Emp_i^{Women}}{Emp_i}$			0.156	0.193	0.097
			(0.215)	(0.212)	(0.223)
$\frac{Emp_i^{Non-Prod}}{Emp_i}$				-0.904***	-0.700***
				(0.206)	(0.219)
$\ln(Investment_i)$					0.061
					(0.064)
$\Delta \ln(Investment_{i,t-1})$					0.040
					(0.025)
$Routineness_i$					-0.213**
					(0.108)
$Automation_i$					-0.085
					(0.058)
$\frac{Materials_i}{Shipments_i}$					0.304
					(0.191)
Constant	0.318***	0.335***	0.144	0.373*	0.557*
	(0.039)	(0.045)	(0.192)	(0.202)	(0.284)
SIC-2 FE	N	N	N	N	N
Obs.	395	395	395	395	395
Adj. R^2	0.075	0.130	0.128	0.161	0.179

Notes: Dependent variable is a dummy variable taking on the value of 1 if an industry's tariff cut was smaller than prescribed by the Swiss formula. OLS estimation of linear probability models. Robust standard errors. *, **, and *** indicate p-values less than 10%, 5% and 1%, respectively.

D Berman et al. (1994) Estimates

Consider groups denoted by $i = 1, ..., I$ and types denoted by $j = 1, ..., J$. In our context, groups are industries and types are production and non-production workers. Let y_{ji} denote the "within" variable of interest for type j within group i. In our context, this is the non-production (NP) pay share within industry i:

$$y_{ji} = \frac{pay_{NP,i}}{pay_i} \tag{B.20}$$

Let y_i denote the "between" shares for group i. In our setting, this is industry i's share of total pay:

$$y_i = \frac{pay_i}{\sum_i pay_i} = \frac{pay_i}{pay}. \tag{B.21}$$

We are interested in how the "aggregate" $y_j = y_{NP}$, in our context the economy-wide non-production pay share, changes over time. More specifically, the variable of interest is

$$\begin{aligned}
\Delta y_j = \Delta y_{NP} &= \Delta \frac{pay_{NP}}{pay} \\
&= \frac{\sum_i pay_{NP,i,t}}{pay_t} - \frac{\sum_i pay_{NP,i,t-1}}{pay_{t-1}} \\
&= \sum_i \frac{pay_{NP,i,t}}{pay_{it}} \frac{pay_{it}}{pay_t} - \sum_i \frac{pay_{NP,i,t-1}}{pay_{i,t-1}} \frac{pay_{i,t-1}}{pay_{t-1}} \\
&= \sum_i y_{NP,it} y_{it} - \sum_i y_{NP,i,t-1} y_{i,t-1}
\end{aligned} \tag{B.22}$$

This can be decomposed into "within" and "between" components

$$\Delta y_{NP} = \sum_i \left(y_{NP,i,t-1} + \frac{1}{2} \Delta y_{NP,i} \right) \Delta y_i + \sum_i \left(y_{i,t-1} + \frac{1}{2} \Delta y_i \right) \Delta y_{NP,i} \tag{B.23}$$

$$= \sum_i \bar{y}_{NP,i} \Delta y_i + \sum_i \bar{y}_i \Delta y_{NP,i}. \tag{B.24}$$

The first summation term is the between component, driven by the between-industry change in pay shares weighted by the industry-specific non-production pay share, with weights based on an average over time. The second summation term is the within component, driven by the change in the within-industry non-production pay shares weighted by industry pay shares, with weights based on an average over time.

Our aim is to obtain estimates of Δy_{NP} caused by tariff changes in order to calculate the share of the observed change in y_{NP} due to tariffs through both the within and between components. In equation (B.24), the $t-1$ variables are observable and we able to estimate Δy_i and $\Delta y_{NP,i}$.

Our main analysis focuses on estimating the change in within-industry non-production pay shares $\Delta y_{NP,i}$ but does not address estimation of the between-industry pay shares. An important complication in estimating these is that they must sum to 1, something which does not constrain estimation of within-industry non-production pay shares). To address this, we estimate industry-level changes in pay and use the set of predicted changes to calculate the predicted change in industry pay shares. That is

$$\begin{aligned}
\Delta y_i &= \Delta \frac{pay_i}{pay} \\
&= \frac{(pay_{it-1} + \Delta pay_i)\, pay_{t-1} - (pay_{t-1} + \Delta pay)\, pay_{i,t-1}}{(pay_{t-1} + \Delta pay)\, pay_{t-1}} \\
&= \frac{(pay_{it-1} + \Delta pay_i)\, pay_{t-1} - (pay_{t-1} + \sum_i \Delta pay_i)\, pay_{i,t-1}}{(pay_{t-1} + \sum_i \Delta pay_i)\, pay_{t-1}} \\
&= \frac{pay_{i,t-1} + \Delta pay_i}{pay_{t-1} + \sum_i \Delta pay_i} - \frac{pay_{i,t-1}}{pay_{t-1}} \\
&= \frac{pay_{i,t-1} + \Delta pay_i}{pay_{t-1} + \sum_i \Delta pay_i} - y_{i,t-1}, \qquad \text{(B.25)}
\end{aligned}$$

which depends only on initial $t-1$ data and estimates of Δpay_i for each industry i.

While we are ultimately interested in Δpay_i, we can recover Δpay_i from estimates of $\Delta \ln(pay_i)$, as

$$\begin{aligned}
\ln(\Delta pay_i) &= \ln\left(\frac{pay_{i,t+1}}{pay_{i,t}}\right) \\
\Rightarrow \exp[\Delta \ln(pay_i)] &= \frac{pay_{i,t+1}}{pay_{i,t}} \qquad \text{(B.26)}
\end{aligned}$$

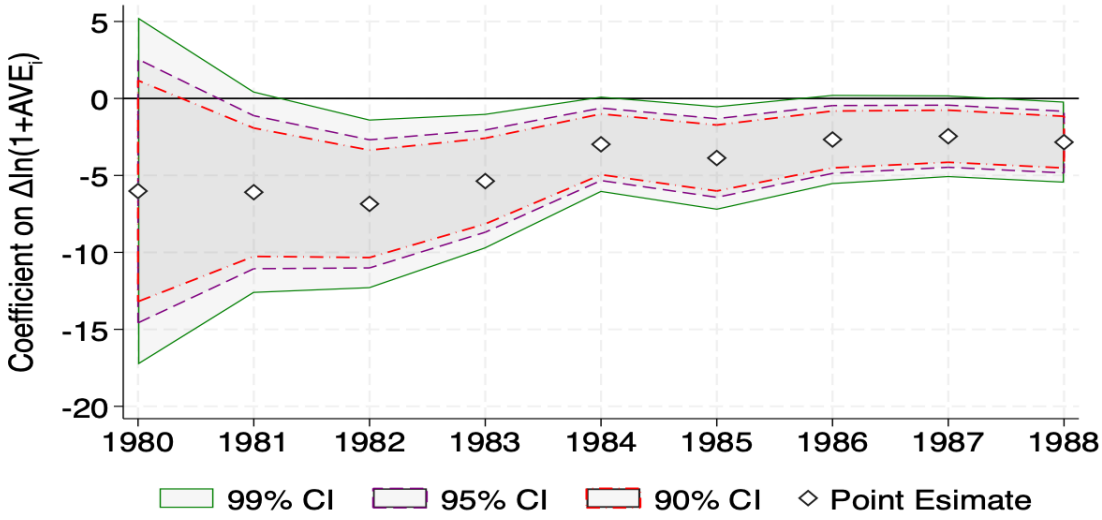
and

$$\begin{aligned}\Delta pay_i &= pay_{i,t+1} - pay_{i,t} \\ &= pay_{i,t}\frac{pay_{i,t+1}}{pay_{i,t}} - pay_{i,t} \\ &= pay_{i,t}\left(\frac{pay_{i,t+1}}{pay_{i,t}} - 1\right) \\ &= pay_{i,t}\left(\exp\left[\Delta \ln\left(pay_i\right)\right] - 1\right). \end{aligned} \tag{B.27}$$

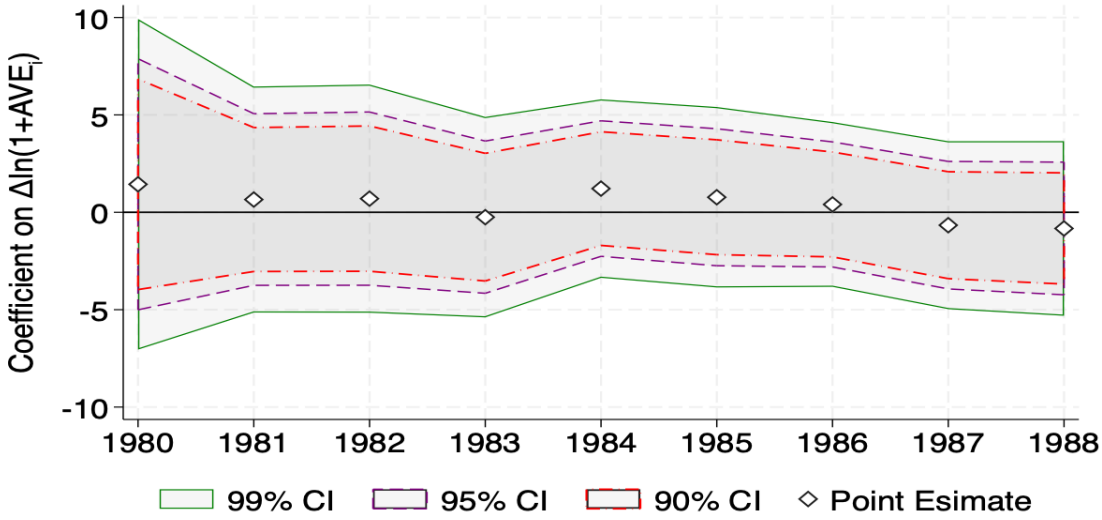
We present the estimated coefficients of our key covariate on both the within industry effects $\Delta y_{NP,i}$ and the between components Δy_i in Figure D.1. As controls we include the full set of controls used throughout the paper.

Consistent with the results of this decomposition presented in Figure 7 in the main text, we find no role for the Tokyo round tariffs affecting the aggregate industry share of pay among NBER manufacturing industries.

Figure D.1: Estimates for Berman et al. (1994) Decomposition



(a) Within Industry Non-Production Pay-Share



(b) Across Industry Pay-Share

Notes: Figure presents estimates of key covariate from equation 1. Dependent variable is within industry non-production pay share in Panel (a), and cross-industry pay-share in Panel (b). Each point is a coefficient from a different regression using a time horizon from 1979-t where t is between 1980 and 1988. Robust standard errors. AVE tariffs from Greenland et al. (2025) and are instrumented with the Swiss formula implied tariff change as in equations 3 and 4. Outcome data taken from the NBER manufacturing database. Data measured at or concorded to the 4-digit SIC87 industry classification.